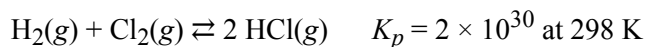
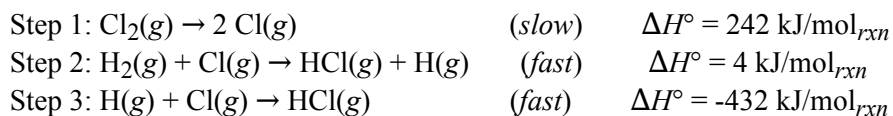


Chapter 10 Part 1: Keq



HCl(g) can be synthesized from H₂(g) and Cl₂(g) as represented above. A student studying the kinetics of the reaction proposes the following mechanism

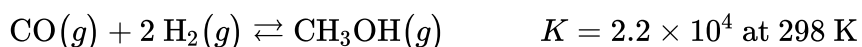
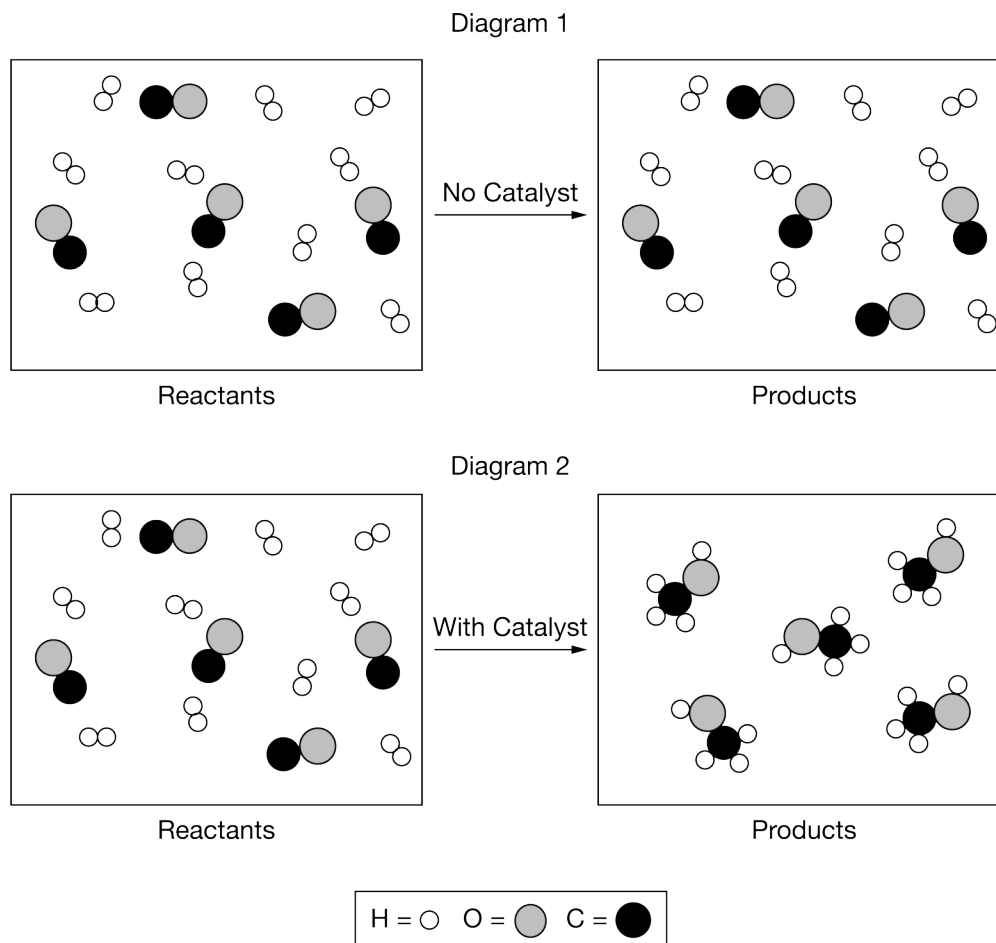


1. Which of the following statements identifies the greatest single reason that the value of K_p for the overall reaction at 298 K has such a large magnitude?
- (A) The activation energy for step 1 of the mechanism is large and positive.
 - (B) The activation energy for step 2 of the mechanism is small and positive.
 - (C) The value of ΔS° for the overall reaction is small and positive.
 - (D) The value of ΔH° for the overall reaction is large and negative.



Chapter 10 Part 1: Keq

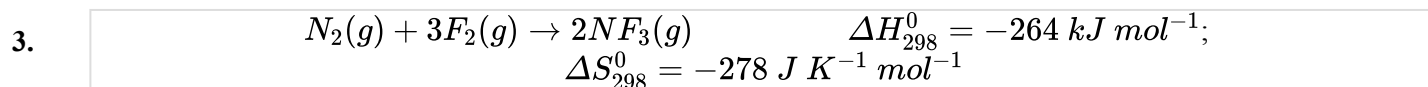
2.



A stoichiometric mixture of $\text{CO}(g)$ and $\text{H}_2(g)$ was allowed to react in two different 2.0 L rigid containers at a constant temperature of 298 K. The reaction is represented by the equation above. Diagram 1 represents the uncatalyzed reaction and diagram 2 represents the catalyzed reaction one hour after the reactants were mixed. Which of the following correctly explains the experimental results represented in the particle diagrams?

- Although the reaction is thermodynamically favorable because $\Delta G^\circ < 0$ based on the value of K ,
- (A) only the catalyzed reaction could proceed in one hour because its reactant molecules had a higher average kinetic energy.
- (B) only the catalyzed reaction could proceed in one hour because it has a lower activation-energy reaction pathway. ✓
- The reaction is not thermodynamically favorable because $\Delta G^\circ > 0$ based on the value of K , but the
- (C) addition of a catalyst improved the orientation of the reactants during collisions, allowing the catalyzed reaction to proceed in one hour.
- (D) The reaction is not thermodynamically favorable because $\Delta G^\circ > 0$ based on the value of K , but the catalyzed reaction could proceed in one hour because it has a lower ΔH and a higher ΔS .

Chapter 10 Part 1: Keq



The following questions relate to the synthesis reaction represented by the chemical equation in the box above.

- Calculate the value of the standard free energy change, ΔG_{298}° , for the reaction.
- Determine the temperature at which the equilibrium constant, K_{eq} , for the reaction is equal to 1.00 . (Assume that ΔH° and ΔS° are independent of temperature.)
- Calculate the standard enthalpy change, ΔH° , that occurs when a 0.256 mol sample of $NF_3(g)$ is formed from $N_2(g)$ and $F_2(g)$ at 1.00 atm and 298 K.

The enthalpy change in a chemical reaction is the difference between energy absorbed in breaking bonds in the reactants and energy released by bond formation in the products.

- How many bonds are formed when two molecules of NF_3 are produced according to the equation in the box above?
- Use both the information in the box above and the table of average bond enthalpies below to calculate the average enthalpy of the F – F bond.

Bond	Average Bond Enthalpy (kJ mol ⁻¹)
N≡N	946
N–F	272
F–F	?

Part A

1 point is earned for correct substitution.

$$\Delta G^{\circ}_{298} = \Delta H^{\circ}_{298} - T\Delta S^{\circ}_{298}$$

1 point is earned for the value of ΔG°_{298} (including kJ or J).

$$= -264 \text{ kJ mol}^{-1} - (298 \text{ K})(-0.278 \text{ kJ mol}^{-1} \text{ K}^{-1})$$

$$= -181 \text{ kJ mol}^{-1}$$

Chapter 10 Part 1: Keq



0	1	2
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The student response earns two of the following points:

1 point is earned for correct substitution.

$$\Delta G^{\circ}298 = \Delta H^{\circ}298 - T\Delta S^{\circ}298$$

1 point is earned for the value of $\Delta G^{\circ}298$ (including kJ or J).

$$= -264 \text{ kJ mol}^{-1} - (298 \text{ K})(-0.278 \text{ kJ mol}^{-1} \text{ K}^{-1})$$

$$= -181 \text{ kJ mol}^{-1}$$

Part B

1 point is earned for indicating that if $Keq = 1$, then $\Delta G^{\circ}T = 0$.

$$\text{When } Keq = 1, \text{ then } \Delta G^{\circ}T = -RT\ln(1) = 0$$

1 point is earned for the answer (including the unit K).

$$\text{If } \Delta G^{\circ}T = 0, \text{ then } 0 = \Delta H^{\circ} - T\Delta S^{\circ} \Rightarrow$$

$$T = \frac{\Delta H^{\circ}_{298}}{\Delta S^{\circ}_{298}}$$

$$T = \frac{-264 \text{ kJ mol}^{-1}}{-0.278 \text{ kJ K}^{-1} \text{ mol}^{-1}} = 950. \text{ K}$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for indicating that if $Keq = 1$, then $\Delta G^{\circ}T = 0$.

$$\text{When } Keq = 1, \text{ then } \Delta G^{\circ}T = -RT\ln(1) = 0$$

1 point is earned for the answer (including the unit K).

$$\text{If } \Delta G^{\circ}T = 0, \text{ then } 0 = \Delta H^{\circ} - T\Delta S^{\circ} \Rightarrow$$

$$T = \frac{\Delta H^{\circ}_{298}}{\Delta S^{\circ}_{298}}$$

Chapter 10 Part 1: Keq

$$T = \frac{-264 \text{ kJ mol}^{-1}}{-0.278 \text{ kJ K}^{-1} \text{ mol}^{-1}} = 950. \text{ K}$$

Part C

1 point is earned for multiplying ΔH°_{298} by the number of moles of NF_3 formed.

1 point is earned for recognizing that 2.00 mol of NF_3 are produced for the reaction as it is written.

1 point is earned for the answer (including kJ or J).

$$0.256 \text{ mol NF}_3(g) \times \frac{-264 \text{ kJ}}{2.00 \text{ mol NF}_3(g)} = -33.8 \text{ kJ}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for multiplying ΔH°_{298} by the number of moles of NF_3 formed.

1 point is earned for recognizing that 2.00 mol of NF_3 are produced for the reaction as it is written.

1 point is earned for the answer (including kJ or J).

$$0.256 \text{ mol NF}_3(g) \times \frac{-264 \text{ kJ}}{2.00 \text{ mol NF}_3(g)} = -33.8 \text{ kJ}$$

Part D

1 point is earned for the correct answer.

There are six N – F bonds formed.



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct answer.

There are six N – F bonds formed.

Part E

1 point is earned for the correct number of bonds in all three compounds multiplied by the average bond enthalpies.

$$\Delta H^\circ_{298} = \sum E_{\text{bonds broken}} - \sum E_{\text{bonds formed}} = -264 \text{ kJ mol}^{-1}$$

Chapter 10 Part 1: Keq

$$\begin{aligned}
 &= [\text{BEN}=\text{N} + (3 \times \text{BEF}-\text{F})] - (6 \times \text{BEN}-\text{F}) \\
 &= [946 \text{ kJ mol}^{-1} + (3 \times \text{BEF}-\text{F})] - 6(272 \text{ kJ mol}^{-1}) \\
 &= -264 \text{ kJ mol}^{-1} \\
 &\Rightarrow 3 \text{ mol BEF}-\text{F} = (-264 - 946 + 1,632) \text{ kJ mol}^{-1} \\
 &1 \text{ point is earned for the answer (including kJ or J).} \\
 &\Rightarrow \text{BEF}-\text{F} = 141 \text{ kJ mol}^{-1}
 \end{aligned}$$

Note: A total of one point is earned if an incorrect number of bonds is substituted in a correct equation and the answer is reasonable (i.e., positive).



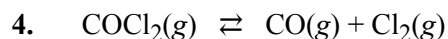
0	1	2
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The student response earns two of the following points:

1 point is earned for the correct number of bonds in all three compounds multiplied by the average bond enthalpies.

$$\begin{aligned}
 \Delta H^\circ_{298} &= \sum E_{\text{bonds broken}} - \sum E_{\text{bonds formed}} = -264 \text{ kJ mol}^{-1} \\
 &= [\text{BEN}=\text{N} + (3 \times \text{BEF}-\text{F})] - (6 \times \text{BEN}-\text{F}) \\
 &= [946 \text{ kJ mol}^{-1} + (3 \times \text{BEF}-\text{F})] - 6(272 \text{ kJ mol}^{-1}) \\
 &= -264 \text{ kJ mol}^{-1} \\
 &\Rightarrow 3 \text{ mol BEF}-\text{F} = (-264 - 946 + 1,632) \text{ kJ mol}^{-1} \\
 &1 \text{ point is earned for the answer (including kJ or J).} \\
 &\Rightarrow \text{BEF}-\text{F} = 141 \text{ kJ mol}^{-1}
 \end{aligned}$$

Note: A total of one point is earned if an incorrect number of bonds is substituted in a correct equation and the answer is reasonable (i.e., positive).



$\text{COCl}_2(\text{g})$ decomposes according to the equation above. When pure $\text{COCl}_2(\text{g})$ is injected into a rigid, previously evacuated flask at 690 K, the pressure in the flask is initially 1.0 atm. After the reaction reaches equilibrium at 690 K, the total pressure in the flask is 1.2 atm. What is the value of K_p for the reaction at 690 K?

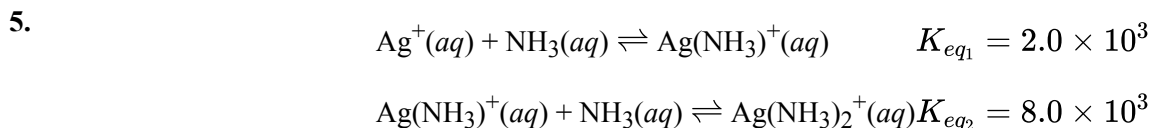
Chapter 10 Part 1: Keq

(A) 0.040

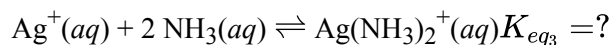
(B) 0.050

(C) 0.80

(D) 1.0



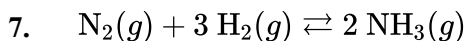
Equal volumes of 0.1 M AgNO₃(aq) and 2.0 M NH₃(aq) are mixed and the reactions represented above occur. Which Ag species will have the highest concentration in the equilibrium system shown below, and why?

(A) Ag⁺(aq), because K_{eq3} = 4(B) Ag⁺(aq), because K_{eq1} < K_{eq2}(C) Ag(NH₃)₂⁺(aq), because K_{eq3} = 1.6 × 10⁷(D) Ag(NH₃)₂⁺(aq), because K_{eq1} < K_{eq2}

Equal volumes of 0.1 M AgNO₃(aq) and 2.0 M NH₃(aq) are mixed and the reactions represented above occur. Which Ag species will have the highest concentration in the equilibrium system shown below, and why?

(A) Ag⁺(aq), because K_{eq3} = 4(B) Ag⁺(aq), because K_{eq1} < K_{eq2}(C) Ag(NH₃)₂⁺(aq), because K_{eq3} = 1.7 × 10⁷(D) Ag(NH₃)₂⁺(aq), because K_{eq1} < K_{eq2}

Chapter 10 Part 1: Keq



$$K = 5.6 \times 10^5 \text{ at } 298 \text{ K}$$

$$\Delta H^\circ_{\text{rxn}} = -91.8 \text{ kJ/mol}_{\text{rxn}}$$

The synthesis of NH_3 is represented by the equation above. Based on the equilibrium constant, K , and $\Delta H^\circ_{\text{rxn}}$ given above, which of the following can best be used to justify that the reaction is thermodynamically favorable at 298 K and constant pressure?

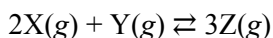
(A) $\Delta G^\circ = -RT \ln K > 0$ because $K \gg 1$

(B) $\Delta G^\circ = -RT \ln K < 0$ because $K \gg 1$ ✓

(C) $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ < 0$ because $\Delta H^\circ < 0$ and $\Delta S^\circ > 0$

(D) $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ > 0$ because $\Delta H^\circ < 0$ and $\Delta S^\circ < 0$

8.



The reaction mixture represented above is at equilibrium at 298 K, and the molar concentrations are $[\text{X}] = 2.0 \text{ M}$, $[\text{Y}] = 0.5 \text{ M}$, and $[\text{Z}] = 4.0 \text{ M}$. What is the value of the equilibrium constant for the reaction at 298 K ?

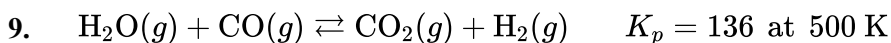
(A) 0.50

(B) 2.0

(C) 4.0

(D) 16

(E) 32 ✓



A vessel initially contains $\text{H}_2\text{O}(g)$ at a partial pressure of 0.30 atm, $\text{CO}(g)$ at a partial pressure of 0.10 atm, $\text{CO}_2(g)$ at a partial pressure of 1.5 atm, and $\text{H}_2(g)$ at a partial pressure of 10. atm at 500 K. Which of the following occurs as the system approaches equilibrium at 500 K?

(A) The partial pressures of $\text{H}_2\text{O}(g)$ and $\text{CO}(g)$ increase because $Q > K_p$. ✓

(B) The partial pressures of $\text{H}_2\text{O}(g)$ and $\text{CO}(g)$ increase because $Q < K_p$.

(C) The partial pressures of $\text{CO}_2(g)$ and $\text{H}_2(g)$ increase because $Q > K_p$.

(D) The partial pressures of $\text{CO}_2(g)$ and $\text{H}_2(g)$ increase because $Q < K_p$.

Chapter 10 Part 1: Keq

- 10.
- | | | |
|-------------|--|---|
| Reaction 1: | $\text{CO}(g) + 3 \text{H}_2(g) \rightleftharpoons \text{CH}_4(g) + \text{H}_2\text{O}(g)$ | $K_1 = \frac{[\text{CH}_4][\text{H}_2\text{O}]}{[\text{CO}][\text{H}_2]^3}$ |
| Reaction 2: | $\text{CO}_2(g) + \text{H}_2(g) \rightleftharpoons \text{CO}(g) + \text{H}_2\text{O}(g)$ | $K_2 = \frac{[\text{CO}][\text{H}_2\text{O}]}{[\text{CO}_2][\text{H}_2]}$ |
| Reaction 3: | $\text{CH}_4(g) + 2 \text{H}_2\text{O}(g) \rightleftharpoons \text{CO}_2(g) + 4 \text{H}_2(g)$ | $K_3 = ?$ |

The chemical equations and equilibrium expressions for two reactions at the same temperature are given above. Based on the information, which of the following expressions can be used to calculate the value of K_3 for reaction 3 at the same temperature?

(A) $K_3 = (-K_1) + (-K_2)$

(B) $K_3 = (-K_1) - (-K_2)$

(C) $K_3 = K_1 \times K_2$

(D) $K_3 = \frac{1}{K_1} \times \frac{1}{K_2}$



- 11.
- $$\text{CO}(g) + 2 \text{H}_2(g) \rightleftharpoons \text{CH}_3\text{OH}(g) \quad \Delta H < 0$$

The synthesis of $\text{CH}_3\text{OH}(g)$ from $\text{CO}(g)$ and $\text{H}_2(g)$ is represented by the equation above. The value of K_c for the reaction at 483 K is 14.5.

A 1.0 mol sample of $\text{CO}(g)$ and a 1.0 mol sample of $\text{H}_2(g)$ are pumped into a rigid, previously evacuated 2.0 L reaction vessel at 483 K. Which of the following is true at equilibrium?

(A) $[\text{H}_2] = 2[\text{CO}]$

(B) $[\text{H}_2] < [\text{CO}]$

(C) $[\text{CO}] = [\text{CH}_3\text{OH}] < [\text{H}_2]$

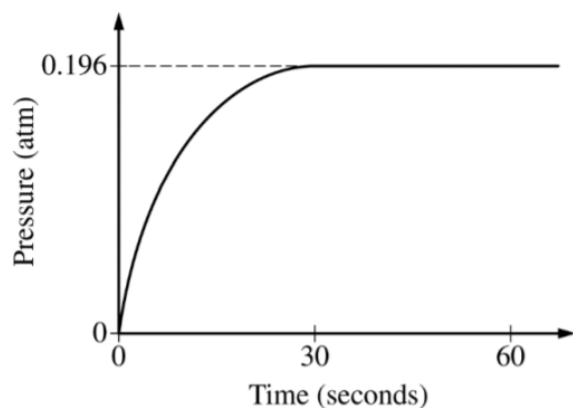
(D) $[\text{CO}] = [\text{CH}_3\text{OH}] = [\text{H}_2]$



Chapter 10 Part 1: Keq

Substance	Lewis Diagram	Boiling Point
CH ₃ OH	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\ddot{\text{O}}-\text{H} \\ \\ \text{H} \end{array}$	338 K
C ₂ H ₅ OH	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\ddot{\text{O}}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	351 K

Equimolar samples of CH₃OH(*l*) and C₂H₅OH(*l*) are placed in separate, previously evacuated, rigid 2.0 L vessels. Each vessel is attached to a pressure gauge, and the temperatures are kept at 300 K. In both vessels, liquid is observed to remain present at the bottom of the container at all times. The change in pressure inside the vessel containing CH₃OH(*l*) is shown below.



12. The value of K_p for the evaporation of CH₃OH(*l*) at 300 K is closest to

(A) 0.04

(B) 0.2

(C) 5

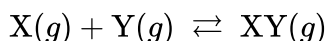
(D) 30



Chapter 10 Part 1: Keq

13.

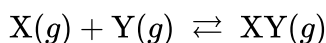
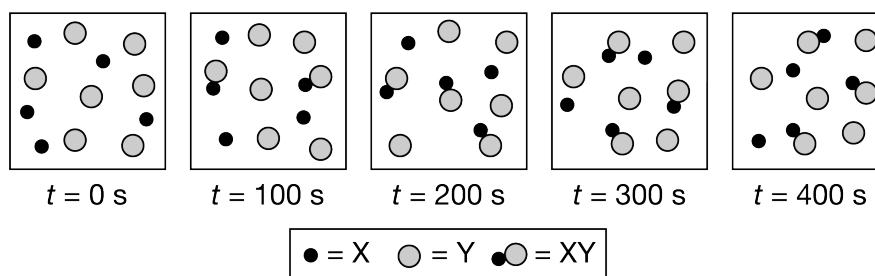
Time (minutes)	[X]	[XY]
0	5.0×10^{-2}	0.0
5	4.1×10^{-2}	9.0×10^{-3}
15	2.9×10^{-2}	2.1×10^{-2}
35	1.0×10^{-2}	4.0×10^{-2}
75	8.0×10^{-3}	4.2×10^{-2}
155	7.0×10^{-3}	4.3×10^{-2}
315	7.0×10^{-3}	4.3×10^{-2}
500	7.0×10^{-3}	4.3×10^{-2}



In an experiment, $\text{X}(g)$ and $\text{Y}(g)$ were combined in a rigid container at constant temperature and allowed to react as shown in the equation above. The table provides the data collected during the experiment. Based on the data, which of the following claims is most likely correct?

- (A) The reaction was about to reach equilibrium 15 minutes after the reactants were combined because the concentrations of X and XY were almost the same.
- (B) The reaction reached equilibrium between 75 minutes and 155 minutes after the reactants were combined because the concentrations of X and XY remained constant. ✓
- (C) The reaction did not reach equilibrium because only 86% of the initial concentration of X was consumed.
- (D) The reaction did not reach equilibrium because initially there was no XY inside the container.

14.

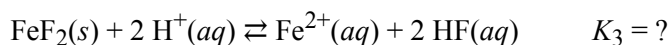
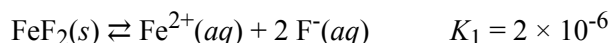


The particle diagram above illustrates the changes that take place when $\text{X}(g)$ and $\text{Y}(g)$ are placed inside a rigid container at constant temperature. Which of the following is a characteristic of a system at equilibrium that is best represented by the particle diagram?

Chapter 10 Part 1: Keq

- (A) The particle diagram shows that initially the reaction proceeds to the right to form products, which is a characteristic of a system at equilibrium.
- (B) The particle diagram shows that after 200 s the rate of the reverse reaction is slower than the rate of the forward reaction, which is a characteristic of a system at equilibrium.
- (C) The particle diagram shows that after 200 s there are no observable changes in the amounts of reactants and products, which is a characteristic of a system at equilibrium. ✓
- (D) The particle diagram shows that between 0 s and 200 s the rates of the forward and reverse reactions are the same, which is a characteristic of a system at equilibrium.

15.



On the basis of the information above, the dissolution of $\text{FeF}_2(s)$ in acidic solution is

- (A) thermodynamically favorable, because $K_2 > 1$
- (B) thermodynamically favorable, because $K_3 > 1$ ✓
- (C) not thermodynamically favorable, because $K_1 < 1$
- (D) not thermodynamically favorable, because $K_3 < 1$

16.

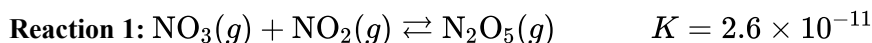
Reaction	Equilibrium Reactions at 298 K	K_{eq}
1	$\text{Br}_2(g) + \text{Cl}_2(g) \rightleftharpoons 2 \text{BrCl}(g)$	10.
2	$\text{N}_2(g) + \text{O}_2(g) \rightleftharpoons 2 \text{NO}(g)$	4.2×10^{-31}

The table above shows data for two reactions carried out in two separate evacuated 1.0-liter rigid containers at constant temperature of 298 K. To each container 0.50 mol of the appropriate reactants was added, and the reaction was allowed to reach equilibrium. Based on this information, which of the following correctly compares the relative concentrations of BrCl and NO present inside their respective containers at equilibrium?

- (A) $[\text{BrCl}]_{eq} = [\text{NO}]_{eq}$ because equimolar mixtures of the reactants were allowed to reach equilibrium at the same constant temperature.
- (B) $[\text{BrCl}]_{eq} > [\text{NO}]_{eq}$ because Br_2 and Cl_2 are larger molecules that can collide more frequently to form products.
- (C) $[\text{BrCl}]_{eq} > [\text{NO}]_{eq}$ because the much larger K_{eq} for reaction 1 means that a much higher concentration of products will be present at equilibrium for reaction 1 compared with reaction 2. ✓
- (D) $[\text{BrCl}]_{eq} < [\text{NO}]_{eq}$ because the much larger K_{eq} for reaction 1 means that hardly any products will be present at equilibrium compared with reaction 2.

Chapter 10 Part 1: Keq

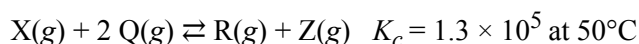
At night, an equilibrium reaction between two different nitrogen compounds generates N_2O_5 in the atmosphere, as represented below.



17. A mixture of NO_3 and a ten-fold excess of NO_2 are placed inside a rigid container at constant temperature and allowed to reach equilibrium. Which of the following provides a correct comparison of the equilibrium concentrations of these chemical species, and why?

- (A) $[\text{N}_2\text{O}_5] \ll [\text{NO}_3]$, because a small K value indicates that the consumption of the reactants is favored at equilibrium.
- (B) $[\text{NO}_2] \ll [\text{N}_2\text{O}_5]$, because a small K value indicates that the consumption of the reactants is favored at equilibrium.
- (C) $[\text{NO}_2] \ll [\text{NO}_3]$, because a small K value indicates that the formation of products is not favored at equilibrium.
- (D) $[\text{N}_2\text{O}_5] \ll [\text{NO}_2]$, because a small K value indicates that the formation of products is not favored at equilibrium. ✓

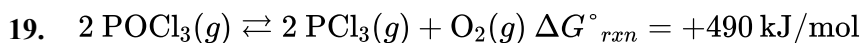
18.



A 1.0 mol sample of $\text{X}(g)$ and a 1.0 mol sample of $\text{Q}(g)$ are introduced into an evacuated, rigid 10.0 L container and allowed to reach equilibrium at 50°C according to the equation above. At equilibrium, which of the following is true about the concentrations of the gases?

- (A) $[\text{R}] = 1/2[\text{Q}]$
- (B) $[\text{Q}] = 1/2[\text{X}]$
- (C) $[\text{R}] = [\text{Z}] > [\text{Q}]$ ✓

- (D) $[\text{X}] = [\text{Q}] = [\text{R}] = [\text{Z}]$



A sample of $\text{POCl}_3(g)$ is placed in a closed, rigid container at 298 K and allowed to reach equilibrium according to the equation above. Based on the value for $\Delta G^\circ_{\text{rxn}} = +490 \text{ kJ/mol}$, which of the following is true?

- (A) $K = e^{-\frac{490,000}{8.314 \times 298}} \ll 1$ and at equilibrium $P_{\text{POCl}_3} \gg P_{\text{PCl}_3}$. ✓

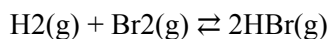
- (B) $K = e^{\frac{8.314 \times 298}{490,000}} > 1$ and at equilibrium $P_{\text{O}_2} > P_{\text{POCl}_3}$.

- (C) $K = e^{-\frac{490}{8.314 \times 298}} < 1$ and at equilibrium $P_{\text{POCl}_3} < P_{\text{PCl}_3}$.

- (D) $K = e^{\frac{8.314 \times 298}{490}} \gg 1$ and at equilibrium $P_{\text{O}_2} \gg P_{\text{POCl}_3}$.

Chapter 10 Part 1: Keq

20.



At a certain temperature, the value of the equilibrium constant, K , for the reaction represented above is 2.0×10^5 . What is the value of K for the reverse reaction at the same temperature?

(A) -2.0×10^{-5}

(B) 5.0×10^{-6}

(C) 2.0×10^{-5}

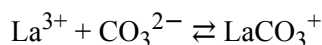
(D) 5.0×10^{-5}

(E) 5.0×10^{-4}

21.

Reaction	K_{eq}
$\text{La}^{3+} + \text{OH}^- + \text{HCO}_3^- \rightleftharpoons \text{LaCO}_3^+ + \text{H}_2\text{O}$	K_1
$\text{HCO}_3^- \rightleftharpoons \text{H}^+ + \text{CO}_3^{2-}$	K_a
$\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$	K_w

Based on the information above, which of the following expressions represents the equilibrium constant, K , for the reaction represented by the equation below?

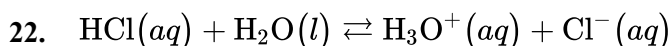


(A) $K = (K_1)(K_a)(K_w)$

(B) $K = (K_1)(K_a) / (K_w)$

(C) $K = (K_1) / ((K_a)(K_w))$

(D) $K = (K_1)(K_w) / (K_a)$

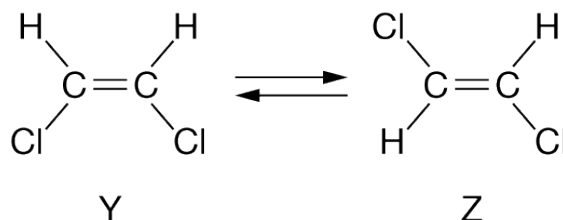


In $1.0 \text{ M HCl}(\text{aq})$, HCl is nearly 100 percent dissociated, as represented by the equation above. Which of the following best helps to explain why, in $0.01 \text{ M HCN}(\text{aq})$, less than 1 percent of HCN is dissociated?

Chapter 10 Part 1: Keq

- (A) The CN^- ion is not very soluble in water, and a solid precipitate would form if more of the HCN dissociated.
- (B) Compared to the $\text{HCl}(aq)$ solution, the concentration of the $\text{HCN}(aq)$ solution is much too dilute to achieve 100 percent dissociation.
- (C) The equilibrium constant for the dissociation of $\text{HCN}(aq)$ is much smaller than that for the dissociation of $\text{HCl}(aq)$. ✓
- (D) $\text{HCN}(aq)$ reacts with water to form a basic solution, and the high concentration of $\text{OH}^-(aq)$ interferes with the dissociation process.

23.

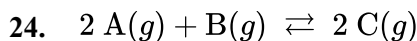


Substance	Y	Z
Initial concentration (M)	2.0	0.0
Equilibrium concentration (M)	1.2	0.80

The diagram above represents the equilibrium between the two isomers of $\text{C}_2\text{H}_2\text{Cl}_2$, and the table provides the data collected in an experiment to determine its equilibrium constant, K_c , at 490 K. In a second experiment done at the same temperature, $[\text{Z}]_{eq} \approx 1.0 M$. Which of the following is the approximate equilibrium concentration of Y in the second experiment, and why?

- (A) $[\text{Y}]_{eq} \approx 1.4 M$ because $[\text{Y}]_{eq} - [\text{Z}]_{eq}$ should be the same for the same reaction.
- (B) $[\text{Y}]_{eq} \approx 1.5 M$ because the ratio $\frac{[\text{Z}]_{eq}}{[\text{Y}]_{eq}}$ should remain constant when the reaction is done at the same temperature. ✓
- (C) $[\text{Y}]_{eq} \approx 1.6 M$ because the ratio $\frac{[\text{Y}]_{initial}}{[\text{Y}]_{eq}}$ should remain constant when the reaction is done at the same temperature.
- (D) $[\text{Y}]_{eq} \approx 2.0 M$ because $([\text{Y}]_{initial} - [\text{Y}]_{eq}) = ([\text{Z}]_{initial} - [\text{Z}]_{eq})$ should be the same for the same reaction.

Chapter 10 Part 1: Keq

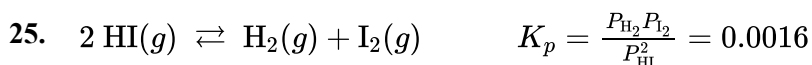


$\text{A}(g)$ and $\text{B}(g)$ react to form $\text{C}(g)$, according to the balanced equation above. In an experiment, a previously evacuated rigid vessel is charged with $\text{A}(g)$, $\text{B}(g)$, and $\text{C}(g)$, each with a concentration of 0.0100 M . The following table shows the concentrations of the gases at equilibrium at a particular temperature.

$[\text{A}]_{eq}$	$[\text{B}]_{eq}$	$[\text{C}]_{eq}$
0.0180	0.0140	0.0020

If the experiment is repeated at a higher temperature at which K_c is larger, which of the following best describes the effect of the temperature change on the concentrations of the gases at equilibrium?

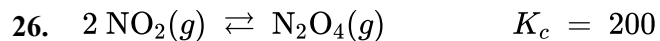
- (A) $[\text{A}]_{eq}$, $[\text{B}]_{eq}$, and $[\text{C}]_{eq}$ will all increase because K_c increased.
- (B) $[\text{A}]_{eq}$ and $[\text{B}]_{eq}$ will remain constant, but $[\text{C}]_{eq}$ will increase because K_c increased.
- (C) There will be a decrease in $[\text{A}]_{eq}$ that will be two times the decrease in $[\text{B}]_{eq}$ because A and B react in a 2-to-1 ratio. ✓
- (D) There will be an increase in $[\text{A}]_{eq}$ that will be two times the increase in $[\text{B}]_{eq}$ because A and B react in a 2-to-1 ratio.



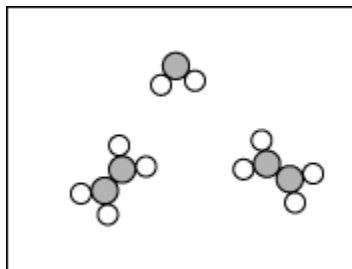
The decomposition of $\text{HI}(g)$ at 298 K is represented by the equilibrium equation above. When $100. \text{ torr}$ of $\text{HI}(g)$ is added to a previously evacuated, rigid container and allowed to reach equilibrium, the partial pressure of $\text{I}_2(g)$ is approximately 3.7 torr . If the initial pressure of $\text{HI}(g)$ is increased to $200. \text{ torr}$ and the process is repeated at the same temperature, which of the following correctly predicts the equilibrium partial pressure of $\text{I}_2(g)$, and why?

- (A) $P_{\text{I}_2} \approx 14 \text{ torr}$, because it is directly proportional to the square of the initial pressure of HI .
- (B) $P_{\text{I}_2} \approx 0.073 \text{ torr}$, because it is inversely proportional to the square of the initial pressure of HI .
- (C) $P_{\text{I}_2} \approx 7.4 \text{ torr}$, because it is directly proportional to the initial pressure of HI . ✓
- (D) $P_{\text{I}_2} \approx 1.9 \text{ torr}$, because it is inversely proportional to the initial pressure of HI .

Chapter 10 Part 1: Keq



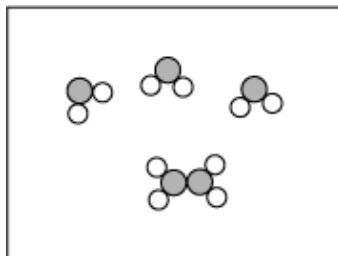
The equilibrium system shown above is represented in the diagram below in which each molecule represents one mole of the substance in a 100. L container at a constant temperature.



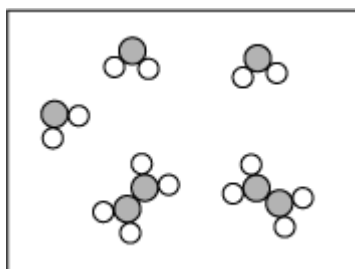
Additional $\text{N}_2\text{O}_4(g)$ is added to the container and allowed to reestablish equilibrium. Which of the following best represents the system after the equilibrium has been reestablished?

Chapter 10 Part 1: Keq

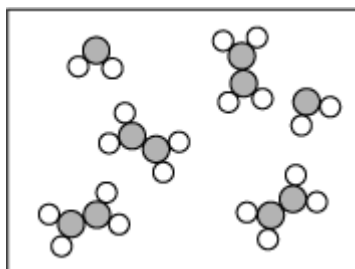
(A)



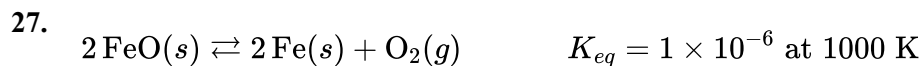
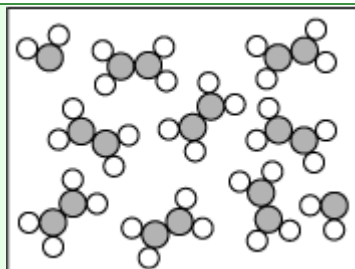
(B)



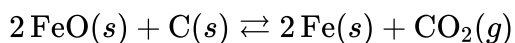
(C)



(D)



The formation of $\text{Fe}(s)$ and $\text{O}_2(g)$ from $\text{FeO}(s)$ is not thermodynamically favorable at room temperature. In an effort to make the process favorable, $\text{C}(s)$ is added to the $\text{FeO}(s)$ at elevated temperatures. Based on the information above, which of the following gives the value of K_{eq} and the sign of ΔG° for the reaction represented by the equation below at 1000 K?



Chapter 10 Part 1: Keq

(A)

K_{eq}	ΔG°
1×10^{-38}	Positive

(B)

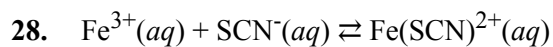
K_{eq}	ΔG°
1×10^{-38}	Negative

(C)

K_{eq}	ΔG°
1×10^{26}	Positive

(D)

K_{eq}	ΔG°
1×10^{26}	Negative



For the reaction represented above, the value of the equilibrium constant, K_{eq} , is 240 at 25°C. From this information, correct deductions about the reaction at 25°C include which of the following?

- I. The reaction is quite rapid.
- II. The product is favored over the reactants at equilibrium.
- III. The reaction is endothermic.

(A) I only

(B) II only

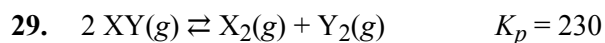
(C) I and II only

(D) II and III only

(E) I, II, and III



Chapter 10 Part 1: Keq



A certain gas, $\text{XY}(g)$, decomposes as represented by the equation above. A sample of each of the three gases is put in a previously evacuated container. The initial partial pressures of the gases are shown in the table below.

Gas	Initial Partial Pressure (atm)
XY	0.010
X_2	0.20
Y_2	2.0

The temperature of the reaction mixture is held constant. In which direction will the reaction proceed?

- (A) The reaction will form more products.
- (B) The reaction will form more reactant. ✓
- (C) The mixture is at equilibrium, so there will be no change.
- (D) It cannot be determined unless the volume of the container is known.

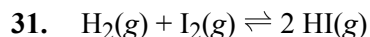
30.

Reaction	K_{eq}
$\text{SO}_2(g) + \frac{1}{2} \text{O}_2(g) \rightleftharpoons \text{SO}_3(g)$	K_1
$2 \text{SO}_3(g) \rightleftharpoons 2 \text{SO}_2(g) + \text{O}_2(g)$	K_2

Which of the following shows the relationship between K_1 and K_2 in the reactions represented above?

- (A) $K_2 = (K_1)^2$
- (B) $K_2 = \sqrt{K_1}$
- (C) $K_2 = \frac{1}{(K_1)^2}$ ✓
- (D) $K_2 = \frac{1}{K_1}$

Chapter 10 Part 1: Keq



$$K_c = 50. \text{ at } 600^\circ\text{C}$$

Equimolar samples of each of three gases, $\text{H}_2(\text{g})$, $\text{I}_2(\text{g})$, and $\text{HI}(\text{g})$, are introduced into a 3.0 L container that is heated to 600°C . According to the information above, which of the following will occur as the system approaches equilibrium?

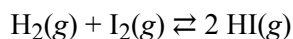
(A) More $\text{H}_2(\text{g})$ will be produced because $Q > K_c$.

(B) More $\text{I}_2(\text{g})$ will be produced because $Q < K_c$.

(C) More $\text{HI}(\text{g})$ will be produced because $Q < K_c$. ✓

(D) No change will occur because $Q = K_c$.

32.



At 450°C , 2.0 moles each of $\text{H}_2(\text{g})$, $\text{I}_2(\text{g})$, and $\text{HI}(\text{g})$ are combined in a 1.0 L rigid container. The value of K_c at 450°C is 50. Which of the following will occur as the system moves toward equilibrium?

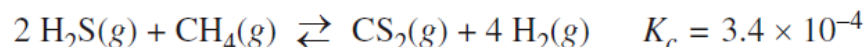
(A) More $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ will form.

(B) More $\text{HI}(\text{g})$ will form. ✓

(C) The total pressure will decrease.

(D) No net reaction will occur, because the number of molecules is the same on both sides of the equation.

33.



A 0.10 mol sample of each of the four species in the reaction represented above is injected into a rigid, previously evacuated 1.0 L container. Which of the following species will have the highest concentration when the system reaches equilibrium?

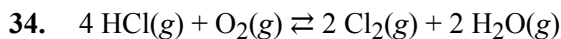
(A) $\text{H}_2\text{S}(\text{g})$ ✓

(B) $\text{CH}_4(\text{g})$

(C) $\text{CS}_2(\text{g})$

(D) $\text{H}_2(\text{g})$

Chapter 10 Part 1: Keq



Equal numbers of moles of HCl and O₂ in a closed system are allowed to reach equilibrium as represented by the equation above. Which of the following must be true at equilibrium?

I. [HCl] must be less than [Cl₂].

II. [O₂] must be greater than [HCl].

III. [Cl₂] must equal [H₂O].

(A) I only

(B) II only

(C) I and III only

(D) II and III only

(E) I, II, and III



35.

$\text{H}_2\text{CO}_3(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{HCO}_3^-(aq) + \text{H}_3\text{O}^+(aq)$	K_1
$\text{HCO}_3^-(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{CO}_3^{2-}(aq) + \text{H}_3\text{O}^+(aq)$	K_2
$\text{NH}_3(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{NH}_4^+(aq) + \text{OH}^-(aq)$	K_3
$2 \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{OH}^-(aq)$	K_4



The table above lists some equilibrium systems and their equilibrium constants. Which of the following identifies the correct mathematical relationship that uses the information to calculate K_5 ?

(A) $K_5 = \frac{K_1 \times K_3}{K_4}$

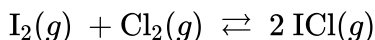
(B) $K_5 = \frac{K_4}{K_1 \times K_3}$

(C) $K_5 = K_1 \times K_3 \times K_4$

(D) $K_5 = \frac{K_1 \times K_3}{K_2}$



Based on the information above, what is the value of K_{eq} for the reaction represented below?



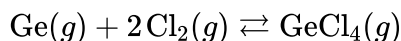
Chapter 10 Part 1: Keq

- (A) 0.01
(B) 10
(C) 20

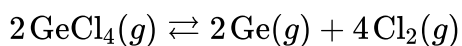
(D) 100



37.



The value of the equilibrium constant for the reaction represented above is 1×10^{10} . What is the value of the equilibrium constant for the following reaction?

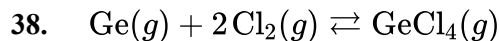


(A) 1×10^{-20}

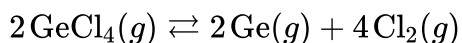
(B) 1×10^{-10}

(C) 1×10^{10}

(D) 1×10^{20}



The value of the equilibrium constant for the reaction represented above is 2.3×10^{11} . What is the value of the equilibrium constant for the following reaction?



(A) 1.9×10^{-23}

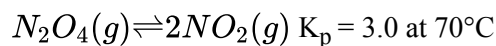
(B) 4.3×10^{-12}

(C) 2.3×10^{11}

(D) 5.3×10^{22}



39.



colorless brown

A mixture of $\text{NO}_2(g)$ and $\text{N}_2\text{O}_4(g)$ is placed in a glass tube and allowed to reach equilibrium at 70°C , as represented above.

If $P_{\text{N}_2\text{O}_4}$ is 0.64 atm when the system is at equilibrium at 70°C , what is P_{NO_2} ?

Chapter 10 Part 1: Keq

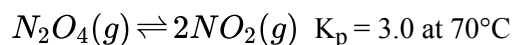
(A) 0.21 atm

(B) 1.4 atm



(C) 1.9 atm

(D) 4.7 atm



colorless brown

A mixture of $\text{NO}_2(g)$ and $\text{N}_2\text{O}_4(g)$ is placed in a glass tube and allowed to reach equilibrium at 70°C , as represented above.

40. If $P_{\text{N}_2\text{O}_4}$ is 1.33 atm when the system is at equilibrium at 70°C , what is P_{NO_2} ?

(A) 0.44 atm

(B) 2.0 atm



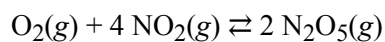
(C) 2.3 atm

(D) 4.0 atm

41.



The equilibrium constant for the gas phase reaction above is 95 at 25°C . What is the value of the equilibrium constant for the following reaction at 25°C ?

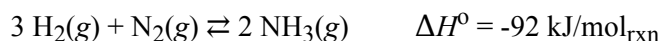
(A) $(95)^2$

(B) 95

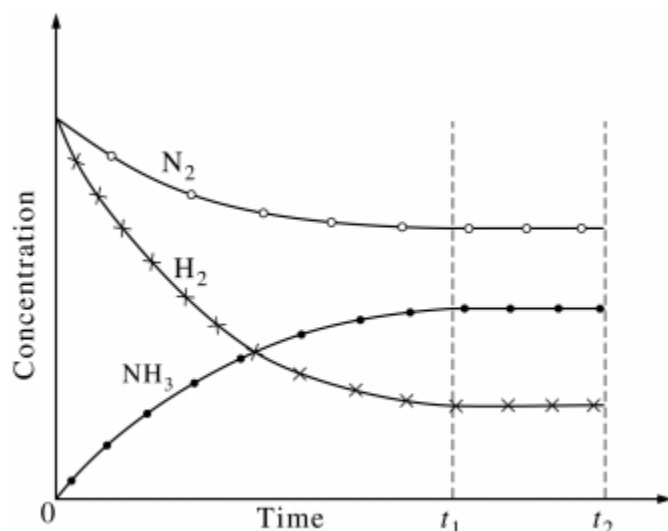
(C) $(95)^{\frac{1}{2}}$ (D) $\frac{1}{95}$ (E) $\frac{1}{(95)^2}$ 

Chapter 10 Part 1: Keq

H₂ gas and N₂ gas were placed in a rigid vessel and allowed to reach equilibrium in the presence of a catalyst according to the following equation.

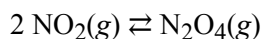


The diagram below shows how the concentrations of H₂, N₂, and NH₃ in this system changed over time.



42. Which of the following was true for the system between time t_1 and time t_2 ?

- (A) The concentration of N₂ decreased.
- (B) The temperature of the system decreased.
- (C) The number of effective collisions between H₂ and N₂ was zero.
- (D) The rates of the forward and reverse reactions were equal. ✓
- (E) The rate of formation of NH₃ molecules was equal to the rate of disappearance of H₂ molecules.



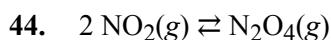
dark brown colorless

The dimerization of NO₂(g), an exothermic process, is represented by the equation above.

43. A 0.0300 mol sample of NO₂(g) is placed in a rigid 1.00 L reaction vessel and allowed to reach equilibrium at a certain temperature. What is the value of K_c at this temperature if 0.00500 mol of N₂O₄(g) is present at equilibrium?

Chapter 10 Part 1: Keq

- (A) 0.0800
(B) 0.250
(C) 4.00
(D) 12.5



dark brown colorless

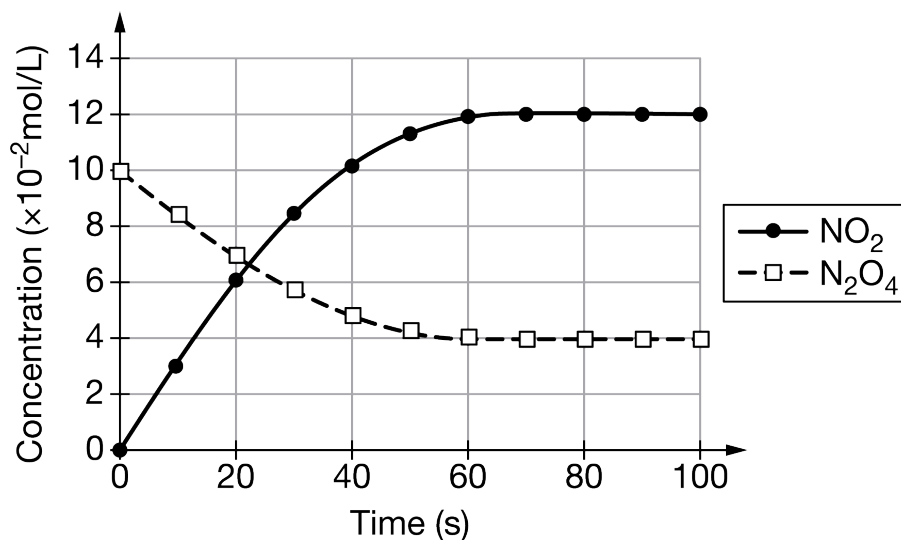
The dimerization of $\text{NO}_2(g)$, an exothermic process, is represented by the equation above.

A 0.0688 mol sample of $\text{NO}_2(g)$ is placed in a rigid 1.00 L reaction vessel and allowed to reach equilibrium at a certain temperature. What is the value of K_c at this temperature if 0.0133 mol of $\text{N}_2\text{O}_4(g)$ is present at equilibrium?

- (A) 0.134
(B) 0.315
(C) 4.32
(D) 7.47



45.

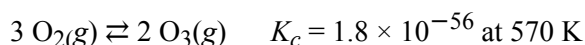


A sample of $\text{N}_2\text{O}_4(g)$ is placed into an evacuated container at 373 K and allowed to undergo the reversible reaction $\text{N}_2\text{O}_4(g) \rightleftharpoons 2 \text{NO}_2(g)$. The concentration of each species is measured over time, and the data are used to make the graph shown above. Which of the following identifies when equilibrium is first reached and provides a correct explanation?

Chapter 10 Part 1: Keq

- (A) At 14 seconds, because $[\text{N}_2\text{O}_4]$ is twice $[\text{NO}_2]$, which implies that the forward and reverse reaction rates are equal.
- (B) At 23 seconds, because $[\text{NO}_2]$ equals $[\text{N}_2\text{O}_4]$, which shows that equal concentrations are present at equilibrium.
- (C) At 40 seconds, because $[\text{NO}_2]$ is twice $[\text{N}_2\text{O}_4]$, which matches the stoichiometry of the balanced chemical equation.
- (D) At 60 seconds, because $[\text{NO}_2]$ and $[\text{N}_2\text{O}_4]$ remain constant, indicating that the forward and reverse reaction rates are equal. ✓

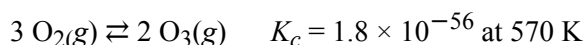
46.



For the system represented above, $[\text{O}_2]$ and $[\text{O}_3]$ initially are 0.150 mol/L and 2.5 mol/L respectively. Which of the following best predicts what will occur as the system approaches equilibrium at 570 K?

- (A) The amount of $\text{O}_3(\text{g})$ will increase, because $Q < K_c$.
- (B) The amount of $\text{O}_3(\text{g})$ will decrease, because $Q < K_c$.
- (C) The amount of $\text{O}_3(\text{g})$ will increase, because $Q > K_c$.
- (D) The amount of $\text{O}_3(\text{g})$ will decrease, because $Q > K_c$. ✓

47.



For the system represented above, $[\text{O}_2]$ and $[\text{O}_3]$ initially are 0.131 mol/L and 0.789 mol/L respectively. Which of the following best predicts what will occur as the system approaches equilibrium at 570 K?

- (A) The amount of $\text{O}_3(\text{g})$ will increase, because $Q < K_c$.
- (B) The amount of $\text{O}_3(\text{g})$ will decrease, because $Q < K_c$.
- (C) The amount of $\text{O}_3(\text{g})$ will increase, because $Q > K_c$.
- (D) The amount of $\text{O}_3(\text{g})$ will decrease, because $Q > K_c$. ✓

Chapter 10 Part 1: Keq

48. Reaction 1: $\text{HOCl}(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{OCl}^-(aq)$ $K_1 = \frac{[\text{H}_3\text{O}^+][\text{OCl}^-]}{[\text{HOCl}]}$
- Reaction 2: $2 \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{OH}^-(aq)$ $K_2 = [\text{H}_3\text{O}^+][\text{OH}^-]$
- Reaction 3: $\text{OCl}^-(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{HOCl}(aq) + \text{OH}^-(aq)$ $K_3 = ?$

Based on the equilibrium constants given above, which of the following gives the correct expression for the equilibrium constant for reaction 3?

(A) $K_3 = \frac{K_2}{K_1}$ ✓

(B) $K_3 = K_1 K_2$

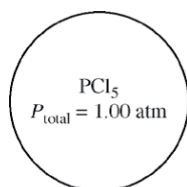
(C) $K_3 = \frac{K_1}{K_2}$

(D) $K_3 = \frac{1}{K_1 K_2}$

Refer to the following.

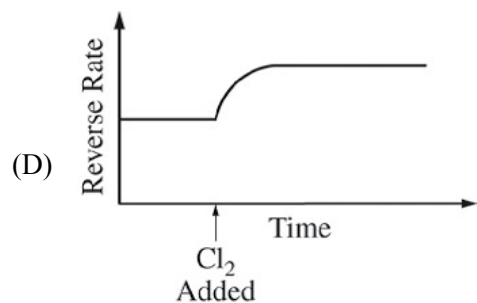
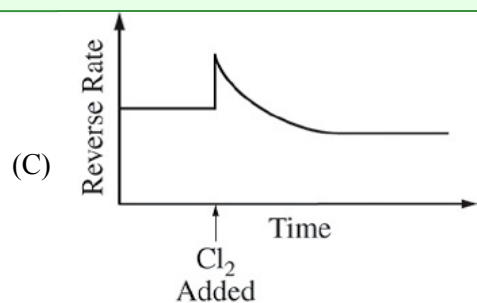
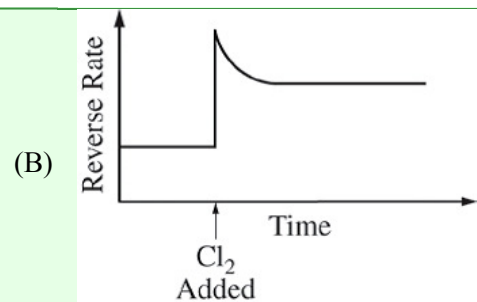
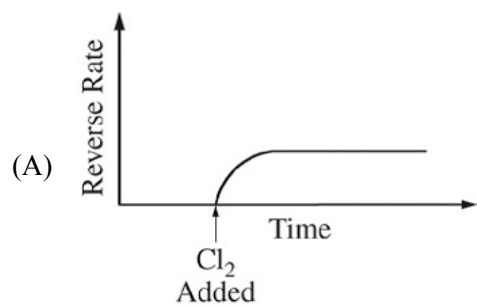


$\text{PCl}_5(g)$ decomposes into $\text{PCl}_3(g)$ and $\text{Cl}_2(g)$ according to the equation above. A pure sample of $\text{PCl}_5(g)$ is placed in a rigid, evacuated 1.00 L container. The initial pressure of the $\text{PCl}_5(g)$ is 1.00 atm. The temperature is held constant until the $\text{PCl}_5(g)$ reaches equilibrium with its decomposition products. The figures below show the initial and equilibrium conditions of the system.



49. As the reaction progresses toward equilibrium, the rate of the forward reaction
- (A) increases until it becomes the same as the reverse reaction rate at equilibrium
- (B) stays constant before and after equilibrium is reached
- (C) decreases to become a constant nonzero rate at equilibrium ✓
- (D) decreases to become zero at equilibrium
50. Additional $\text{Cl}_2(g)$ is injected into the system at equilibrium. Which of the following graphs best shows the rate of the reverse reaction as a function of time? (Assume that the time for injection and mixing of the additional $\text{Cl}_2(g)$ is negligible.)

Chapter 10 Part 1: Keq



51. Which of the following statements about K_p , the equilibrium constant for the reaction, is correct?

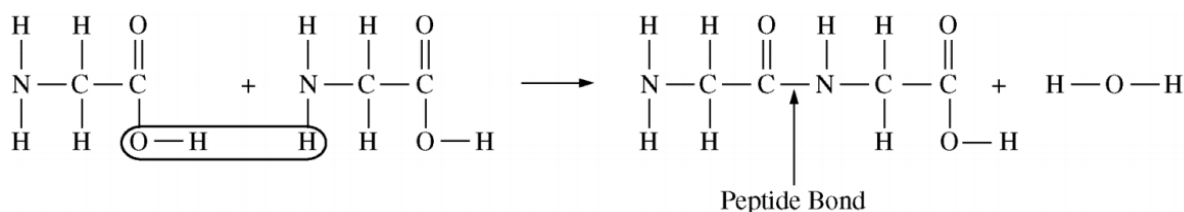
(A) $K_p > 1$

(B) $K_p < 1$

(C) $K_p = 1$

(D) It cannot be determined whether $K_p > 1$, $K_p < 1$, or $K_p = 1$ without additional information.

Chapter 10 Part 1: Keq



Two molecules of the amino acid glycine join through the formation of a peptide bond, as shown above. The thermodynamic data for the reaction are listed in the following table.

ΔG_{298}°	ΔH_{298}°	ΔS_{298}°
+15 kJ/mol _{rxn}	+12 kJ/mol _{rxn}	-10 J/(K·mol _{rxn})

52. Based on the thermodynamic data, which of the following is true at 298 K?

(A) $K_{eq} = 0$

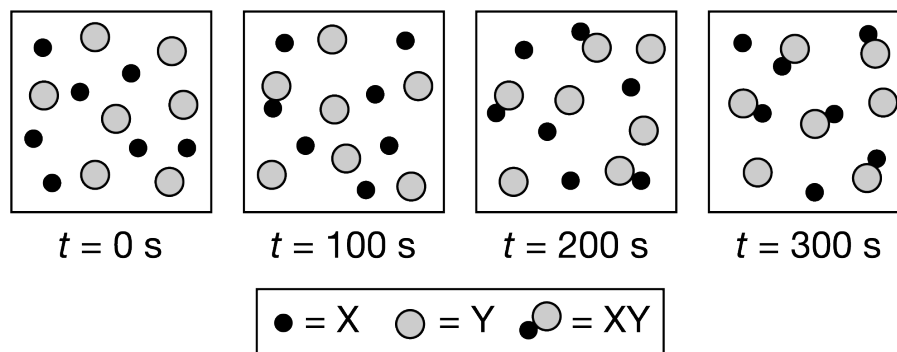
(B) $0 < K_{eq} < 1$

(C) $K_{eq} = 1$

(D) $K_{eq} > 1$



53.

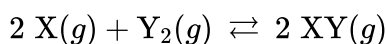
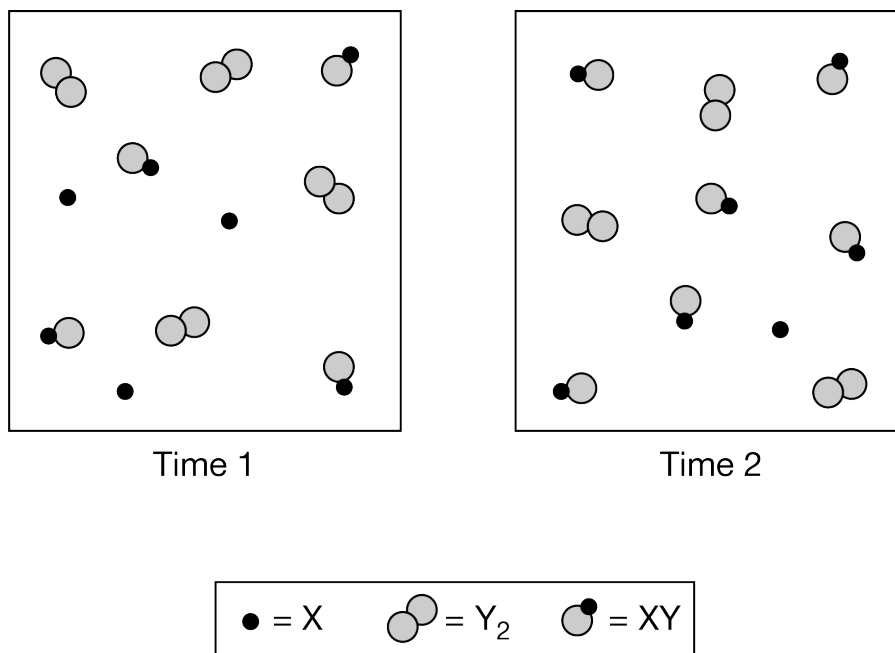


An equimolar mixture of $X(g)$ and $Y(g)$ is placed inside a rigid container at constant temperature. The particle diagram above represents the changes that occur over time. Based on the particle diagram, which of the following best predicts whether or not the system has reached equilibrium by 300 s?

Chapter 10 Part 1: Keq

- (A) It is not possible to determine that the system has reached equilibrium by 300 s because the stoichiometry of the reaction is not known.
- (B) It is not possible to determine that the system has reached equilibrium by 300 s because the amounts of X, Y, and XY have continued to change. ✓
- (C) The system has reached equilibrium by 300 s because the rate of formation of XY is constant.
- (D) The system has reached equilibrium by 300 s because the rates of consumption of X and Y are equal.

54.



A reversible reaction is represented by the equation above. The amounts of reactants and products at time 1 are shown in the particle diagram on the left. The particle diagram on the right shows the amounts of reactants and products at time 2. Based on the diagrams, what can be inferred about the relative rates of the forward and reverse reactions between time 1 and time 2 ?

- (A) Nothing can be inferred because the total number of X and Y atoms is the same in each diagram.
- (B) Nothing can be inferred because the temperature of the system may have been changed.
- (C) The rate of the reverse reaction is greater than the rate of the forward reaction.
- (D) The rate of the forward reaction is greater than the rate of the reverse reaction. ✓

55.



At a certain point in time, a 1.00 L rigid reaction vessel contains 1.5 mol of $\text{PCl}_3(g)$, 1.0 mol of $\text{Cl}_2(g)$, and 2.5 mol of $\text{PCl}_5(g)$. Which of the following describes how the measured pressure in the reaction vessel will change and why it will change that way as the reaction system approaches equilibrium at constant temperature?

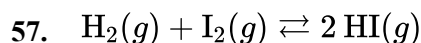
Chapter 10 Part 1: Keq

- (A) The pressure will increase because $Q < K_c$.
(B) The pressure will increase because $Q > K_c$.
(C) The pressure will decrease because $Q < K_c$. ✓
(D) The pressure will decrease because $Q > K_c$.



Equal volumes of 0.1 M $\text{AgNO}_3(aq)$ and 0.4 M $\text{NH}_3(aq)$ are mixed and allowed to reach the equilibrium represented above. Which of the following correctly lists the equilibrium concentrations of the $\text{Ag}^+(aq)$, $\text{NH}_3(aq)$, and $[\text{Ag}(\text{NH}_3)_2]^+(aq)$ in order from least to greatest?

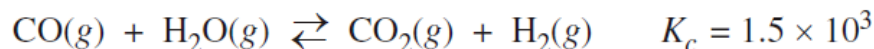
- (A) $[\text{Ag}^+]_{eq} < [\text{NH}_3]_{eq} < [[\text{Ag}(\text{NH}_3)_2]^+]_{eq}$
(B) $[\text{NH}_3]_{eq} < [\text{Ag}^+]_{eq} < [[\text{Ag}(\text{NH}_3)_2]^+]_{eq}$
(C) $[[\text{Ag}(\text{NH}_3)_2]^+]_{eq} < [\text{Ag}^+]_{eq} < [\text{NH}_3]_{eq}$
(D) $[\text{Ag}^+]_{eq} < [[\text{Ag}(\text{NH}_3)_2]^+]_{eq} < [\text{NH}_3]_{eq}$ ✓



Hydrogen gas reacts with iodine gas at constant temperature in a sealed rigid container. The gases are allowed to reach equilibrium according to the equation above. Which of the following best describes what will happen to the reaction immediately after additional iodine gas is added to the system?

- (A) The rates of both the forward and reverse reactions decrease.
(B) The rates of both the forward and reverse reactions do not change.
(C) The rate of the forward reaction becomes greater than the rate of the reverse reaction. ✓
(D) The rate of the forward reaction becomes less than the rate of the reverse reaction.

58.

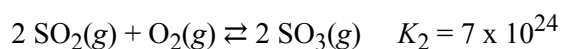


A 2.0 mol sample of $\text{CO}(g)$ and a 2.0 mol sample of $\text{H}_2\text{O}(g)$ are introduced into a previously evacuated 100. L rigid container, and the temperature is held constant as the reaction represented above reaches equilibrium. Which of the following is true at equilibrium?

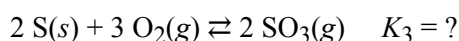
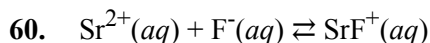
- (A) $[\text{H}_2\text{O}] > [\text{CO}]$ and $[\text{CO}_2] > [\text{H}_2]$
(B) $[\text{H}_2\text{O}] > [\text{H}_2]$
(C) $[\text{CO}_2] > [\text{CO}]$ ✓
(D) $[\text{CO}] = [\text{H}_2\text{O}] = [\text{CO}_2] = [\text{H}_2]$

Chapter 10 Part 1: Keq

59.



Given the value of the equilibrium constants K_1 and K_2 for the reactions represented above, what is the value of the equilibrium constant, K_3 , for the following reaction?

(A) 1×10^{130} (B) 3×10^{80} (C) 1×10^{65} (D) 2×10^{40} (E) 7×10^{24} 

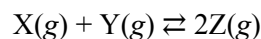
At 25°C, the equilibrium constant for the reaction represented above has a value of 1.3. At 50°C, the value of the equilibrium constant is less than 1.3. Based on this information, which of the following must be correct?

(A) The reaction rate decreases as the temperature is increased.

(B) The reaction is thermodynamically favorable only at temperatures above 25°C.

(C) At 25°C, ΔG° for the reaction is positive.(D) At 25°C, ΔS° for the reaction is positive.(E) At 25°C, ΔH° for the reaction is negative.

61.



When 4.00 mol each of X(g) and Y(g) are placed in a 1.00 L vessel and allowed to react at constant temperature according to the equation above, 6.00 mol of Z(g) is produced. What is the value of the equilibrium constant, K_c ?

(A) 3

(B) 6

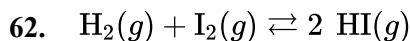
(C) 8

(D) 16

(E) 36



Chapter 10 Part 1: Keq



A 1.0 mol sample of $\text{H}_2(g)$ and a 1.0 mol sample of $\text{I}_2(g)$ are placed in a previously evacuated 1.0 L container and allowed to reach the equilibrium shown above. If the equilibrium concentration of $\text{HI}(g)$ is 0.40 M, which of the following is the value of the equilibrium constant?

(A) 0.16

(B) 0.25

(C) 0.44

(D) 1.0



63. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



When heated strongly, solid calcium carbonate decomposes to produce solid calcium oxide and carbon dioxide gas, as represented by the equation above. A 2.0 mol sample of $\text{CaCO}_3(s)$ is placed in a rigid 100. L reaction vessel from which all the air has been evacuated. The vessel is heated to 898°C at which time the pressure of $\text{CO}_2(g)$ in the vessel is constant at 1.00 atm, while some $\text{CaCO}_3(s)$ remains in the vessel.

- (a) Calculate the number of moles of $\text{CO}_2(g)$ present in the vessel at equilibrium.
- (b) Write the expression for K_p , the equilibrium constant for the reaction, and determine its value at 898°C .
- (c) The experiment was repeated, but this time starting with a 4.0 mol sample of $\text{CaCO}_3(s)$. On the following graph, draw a curve showing how the pressure of $\text{CO}_2(g)$ would change over time as the vessel is heated to 898°C and equilibrium is established.



Part (a)

Chapter 10 Part 1: Keq

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response includes both the following set up and calculation:

$$n = \frac{PV}{RT} = \frac{(1.00 \text{ atm})(100. \text{ L})}{(0.0821 \text{ L} \cdot \text{atm} \cdot \text{mol}^{-1} \cdot \text{K}^{-1})(1171 \text{ K})} = 1.04 \text{ mol}$$

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response includes both of the following two criteria:

- ☐ $K_p = P_{\text{CO}_2}$
- ☐ $K_p = 1.00 \text{ atm}$

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response

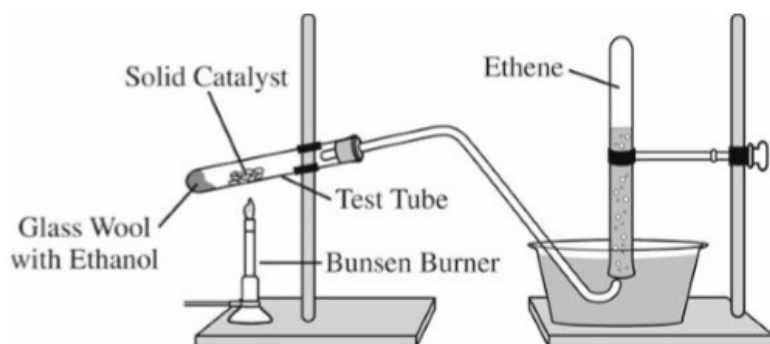


0	1
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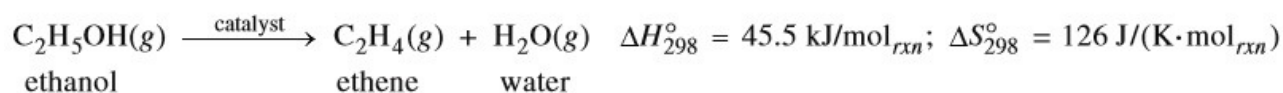
The response shows a graph that starts at the origin, rises to right, and levels off (slope = 0) at 1.00 atm

Chapter 10 Part 1: Keq

64.



Ethene, $\text{C}_2\text{H}_4(\text{g})$ (molar mass 28.1 g/mol), may be prepared by the dehydration of ethanol, $\text{C}_2\text{H}_5\text{OH}(\text{g})$ (molar mass 46.1 g/mol), using a solid catalyst. A setup for the lab synthesis is shown in the diagram above. The equation for the dehydration reaction is given below.



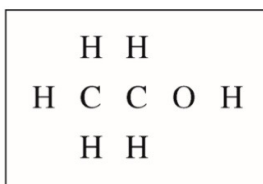
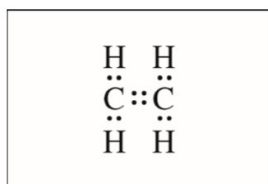
A student added a 0.200 g sample of $\text{C}_2\text{H}_5\text{OH}(\text{l})$ to a test tube using the setup shown above. The student heated the test tube gently with a Bunsen burner until all of the $\text{C}_2\text{H}_5\text{OH}(\text{l})$ evaporated and gas generation stopped. When the reaction stopped, the volume of collected gas was 0.0854 L at 0.822 atm and 305 K. (The vapor pressure of water at 305 K is 35.7 torr.)

- a. Calculate the number of moles of $\text{C}_2\text{H}_4(\text{g})$
 - i. that are actually produced in the experiment and measured in the gas collection tube and
 - ii. that would be produced if the dehydration reaction went to completion.

- b. Calculate the percent yield of $\text{C}_2\text{H}_4(\text{g})$ in the experiment.

Because the dehydration reaction is not observed to occur at 298 K, the student claims that the reaction has an equilibrium constant less than 1.00 at 298 K.

- c. Do the thermodynamic data for the reaction support the student's claim? Justify your answer, including a calculation of ΔG_{298}° for the reaction.
- d. The Lewis electron-dot diagram for C_2H_4 is shown below in the box on the left. In the box on the right, complete the Lewis electron-dot diagram for $\text{C}_2\text{H}_5\text{OH}$ by drawing in all of the electron pairs.



Chapter 10 Part 1: Keq

- e. What is the approximate value of the C–O–H bond angle in the ethanol molecule?
- f. During the dehydration experiment, $\text{C}_2\text{H}_4(g)$ and unreacted $\text{C}_2\text{H}_5\text{OH}(g)$ passed through the tube into the water. The C_2H_4 was quantitatively collected as a gas, but the unreacted $\text{C}_2\text{H}_5\text{OH}$ was not. Explain this observation in terms of the intermolecular forces between water and each of the two gases.

Part A

- 1 point is earned for the calculation of the pressure of the dry ethene.

$$35.7 \text{ torr} \times \frac{1 \text{ atm}}{760 \text{ torr}} = 0.0470 \text{ atm}$$

$$P_{\text{ethene}} = P_{\text{total}} - P_{\text{water}} = 0.822 \text{ atm} - 0.0470 \text{ atm} = 0.775 \text{ atm}$$

1 point is earned for the correct number of moles of ethene gas.

$$n = \frac{PV}{RT} = \frac{(0.775 \text{ atm})(0.0854 \text{ L})}{(0.08206 \text{ L atm mol}^{-1}\text{K}^{-1})(305 \text{ K})} = 0.00264 \text{ mol}$$

- 1 point is earned for the correct number of moles of ethene produced.

$$0.200 \text{ g C}_2\text{H}_5\text{OH} \times \frac{1 \text{ mol C}_2\text{H}_5\text{OH}}{46.1 \text{ g C}_2\text{H}_5\text{OH}} \times \frac{1 \text{ mol C}_2\text{H}_4}{1 \text{ mol C}_2\text{H}_5\text{OH}}$$

$$= 0.00434 \text{ mol C}_2\text{H}_4 \text{ produced}$$



0	1	2	3
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The student response earns three of the following points:

- 1 point is earned for the calculation of the pressure of the dry ethene.

$$35.7 \text{ torr} \times \frac{1 \text{ atm}}{760 \text{ torr}} = 0.0470 \text{ atm}$$

$$P_{\text{ethene}} = P_{\text{total}} - P_{\text{water}} = 0.822 \text{ atm} - 0.0470 \text{ atm} = 0.775 \text{ atm}$$

1 point is earned for the correct number of moles of ethene gas.

$$n = \frac{PV}{RT} = \frac{(0.775 \text{ atm})(0.0854 \text{ L})}{(0.08206 \text{ L atm mol}^{-1}\text{K}^{-1})(305 \text{ K})} = 0.00264 \text{ mol}$$

- 1 point is earned for the correct number of moles of ethene produced.

$$0.200 \text{ g C}_2\text{H}_5\text{OH} \times \frac{1 \text{ mol C}_2\text{H}_5\text{OH}}{46.1 \text{ g C}_2\text{H}_5\text{OH}} \times \frac{1 \text{ mol C}_2\text{H}_4}{1 \text{ mol C}_2\text{H}_5\text{OH}}$$

$$= 0.00434 \text{ mol C}_2\text{H}_4 \text{ produced}$$

Part B

Chapter 10 Part 1: Keq

1 point is earned for the correct percent yield.

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{maximum possible yield}} \times 100 = \frac{0.00264 \text{ mol}}{0.00434 \text{ mol}} \times 100 = 60.8\%$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct percent yield.

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{maximum possible yield}} \times 100 = \frac{0.00264 \text{ mol}}{0.00434 \text{ mol}} \times 100 = 60.8\%$$

Part C

1 point is earned for the correct calculation of ΔG° .

Yes, the data support the student's claim.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$= 45.5 \text{ kJ/molrxn} - (298 \text{ K})(0.126 \text{ kJ/K} \cdot \text{molrxn}) = 8.0 \text{ kJ/molrxn}$$

1 point is earned for a valid justification.

Because $\Delta G^\circ > 0$, the value of $K_p = e^{(-\frac{\Delta G^\circ}{RT})} < 1.00$.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct calculation of ΔG° .

Yes, the data support the student's claim.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$= 45.5 \text{ kJ/molrxn} - (298 \text{ K})(0.126 \text{ kJ/K} \cdot \text{molrxn}) = 8.0 \text{ kJ/molrxn}$$

1 point is earned for a valid justification.

Because $\Delta G^\circ > 0$, the value of $K_p = e^{(-\frac{\Delta G^\circ}{RT})} < 1.00$.

Part D

Chapter 10 Part 1: Keq

1 point is earned for a correct diagram.

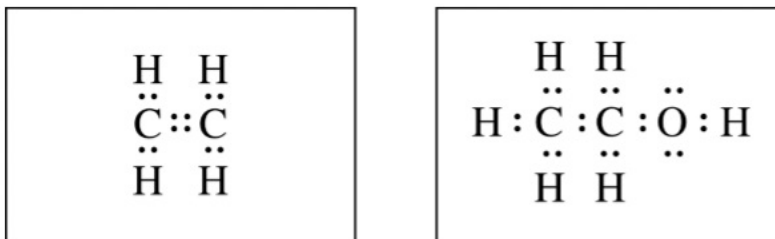


Diagram should include all bonding pairs plus two nonbonding pairs on the O atom. (A line may be used to represent an electron pair.)



0

1

The student response earns one of the following points:

1 point is earned for a correct diagram.

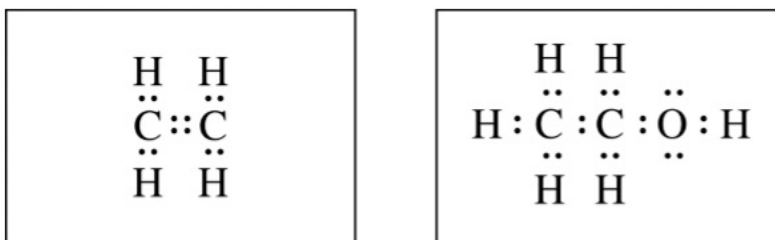


Diagram should include all bonding pairs plus two nonbonding pairs on the O atom. (A line may be used to represent an electron pair.)

Part E

1 point is earned for an angle from 100° to 115°.

The bond angle is approximately 109°.



0

1

The student response earns one of the following points:

1 point is earned for an angle from 100° to 115°.

The bond angle is approximately 109°.

Chapter 10 Part 1: Keq

Part F

1 point is earned for comparing the solubility of ethene in water with the solubility of ethanol in water in terms of differences in polarity.

1 point is earned for describing the intermolecular forces between ethene and water as weak dipole/induced dipole forces and attributing the solubility of ethanol in water to the hydrogen bonds formed between ethanol molecules and water molecules.

Ethene is only slightly soluble in water because the weak dipole/induced dipole intermolecular attractions between nonpolar ethene molecules and polar water molecules are weaker than the hydrogen bonds between water molecules. Ethanol molecules are soluble in water because they are polar and form hydrogen bonds with water molecules as they dissolve.



0	1	2
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The student response earns two of the following points:

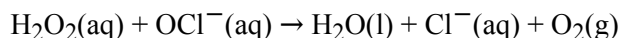
1 point is earned for comparing the solubility of ethene in water with the solubility of ethanol in water in terms of differences in polarity.

1 point is earned for describing the intermolecular forces between ethene and water as weak dipole/induced dipole forces and attributing the solubility of ethanol in water to the hydrogen bonds formed between ethanol molecules and water molecules.

Ethene is only slightly soluble in water because the weak dipole/induced dipole intermolecular attractions between nonpolar ethene molecules and polar water molecules are weaker than the hydrogen bonds between water molecules. Ethanol molecules are soluble in water because they are polar and form hydrogen bonds with water molecules as they dissolve.

Chapter 10 Part 1: Keq

65.



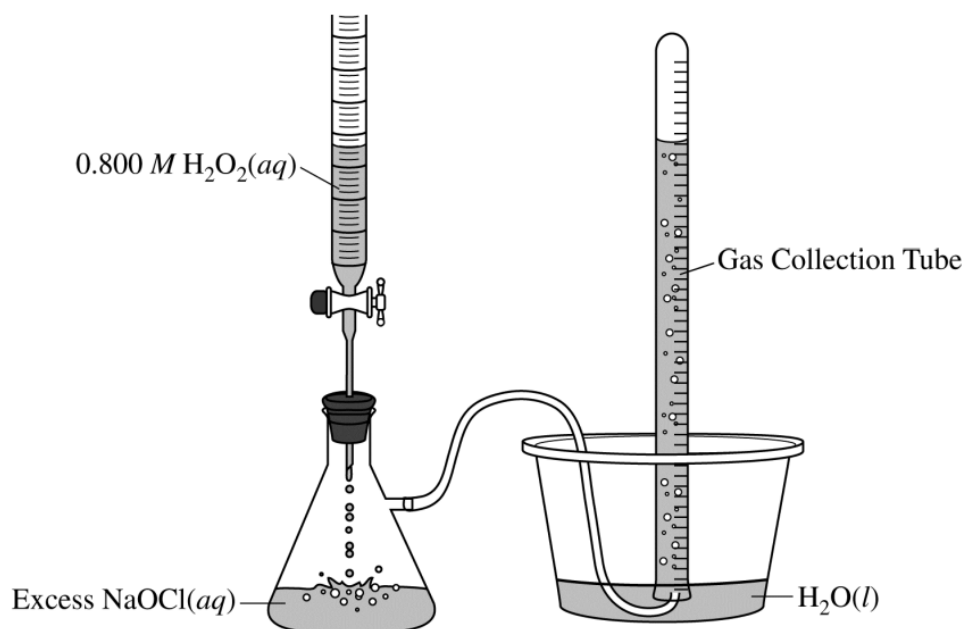
A student investigates the reaction between $\text{H}_2\text{O}_2(\text{aq})$ and $\text{NaOCl}(\text{aq})$, which is represented by the net-ionic equation shown above.

- a. Is the reaction represented above a redox reaction? Justify your answer.

To better understand the reaction, the student looks up thermodynamic data for the reaction. For the reaction represented above, ΔG_{298}° is $-197 \text{ kJ/mol}_{\text{rxn}}$ represented above, the value of and the value of ΔS_{298}° is $144 \text{ J/(K mol}_{\text{rxn}})$

- b. Calculate the value of ΔH_{298}° for the reaction in $\text{kJ/mol}_{\text{rxn}}$.
- c. Does the temperature inside the flask increase, decrease, or remain the same as the reaction proceeds? Justify your answer.
- d. Calculate the value of the equilibrium constant, K , for the reaction at 298 K

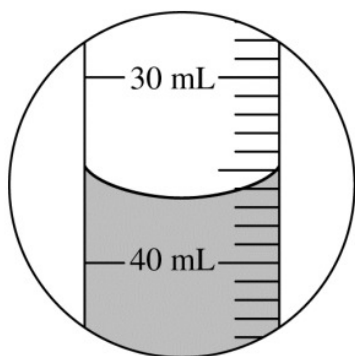
The student decides to produce 40.0 mL of $\text{O}_2(\text{g})$ at a pressure of 0.988 atm and a temperature of 298 K using the reaction represented above. The student uses the equipment shown below. The student sets up a 250 mL Erlenmeyer flask fitted with a one-hole stopper. The flask is connected to a 50 mL gas-collection tube that initially is completely filled with water.



- e. Calculate the volume of 0.800 M $\text{H}_2\text{O}_2(\text{aq})$ that the student should add to excess $\text{NaOCl}(\text{aq})$ to produce 40.0 mL of $\text{O}_2(\text{g})$ at 0.988 atm and 298 K.
- f. The student added the amount of $\text{H}_2\text{O}_2(\text{aq})$ calculated in part (e) to excess $\text{NaOCl}(\text{aq})$. However, instead

Chapter 10 Part 1: Keq

of producing 40.0 mL of $O_2(g)$, the volume indicated in the diagram below was produced.



- i. Based on the diagram above, what volume of gas was produced?
 - ii. Assuming that all the gas in the tube is $O_2(g)$, calculate the percent yield of $O_2(g)$.
 - iii. Is the assumption that all the gas in the tube is $O_2(g)$ correct? Explain.
- g. To account for the percent yield being less than 100 percent, the student claims that the reaction reached equilibrium before the expected amount of $O_2(g)$ was produced. Considering your answer to part (d) above, do you agree or disagree with the student's claim? Justify your answer.

Part A

1 point(s) maximum

1 point is earned for the correct answer along with a valid justification.

The reaction is a redox reaction because the oxidation numbers of some atoms changed during the reaction (both oxygen and chlorine undergo changes in oxidation number).



0

1

Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for the correct answer along with a valid justification.

The reaction is a redox reaction because the oxidation numbers of some atoms changed during the reaction (both oxygen and chlorine undergo changes in oxidation number).

Part B

1 point(s) maximum

Chapter 10 Part 1: Keq

1 point is earned for the correct calculation of ΔH° .

$$\begin{aligned}\Delta H^\circ &= \Delta G^\circ + T\Delta S^\circ \\ &= -197 \text{ kJ/mol}_{\text{rxn}} + (298 \text{ K}) \left(\frac{144 \text{ J}}{\text{K} \cdot \text{mol}_{\text{rxn}}} \right) \left(\frac{1 \text{ kJ}}{1000 \text{ J}} \right) \\ &= -154 \text{ kJ/mol}_{\text{rxn}}\end{aligned}$$



0	1
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Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for the correct calculation of ΔH° .

$$\begin{aligned}\Delta H^\circ &= \Delta G^\circ + T\Delta S^\circ \\ &= -197 \text{ kJ/mol}_{\text{rxn}} + (298 \text{ K}) \left(\frac{144 \text{ J}}{\text{K} \cdot \text{mol}_{\text{rxn}}} \right) \left(\frac{1 \text{ kJ}}{1000 \text{ J}} \right) \\ &= -154 \text{ kJ/mol}_{\text{rxn}}\end{aligned}$$

Part C**1 point(s) maximum**

1 point is earned for indicating an increase in temperature with a valid justification.

The temperature increases because the reaction is exothermic ($\Delta H^\circ < 0$).



0	1
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Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for indicating an increase in temperature with a valid justification.

The temperature increases because the reaction is exothermic ($\Delta H^\circ < 0$).

Part D**1 point(s) maximum**

Chapter 10 Part 1: Keq

1 point is earned for the correct value of K with evidence of calculation.

$$\Delta G^\circ = -RT \ln K$$

$$K = e^{\frac{-\Delta G^\circ}{RT}} = e^{\frac{-(-197,000 \text{ J/mol})}{(8.314 \text{ J/(mol}\cdot\text{K)})(298 \text{ K})}} = e^{79.5} = 3 \times 10^{34}$$



0

1

Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for the correct value of K with evidence of calculation.

$$\Delta G^\circ = -RT \ln K$$

$$K = e^{\frac{-\Delta G^\circ}{RT}} = e^{\frac{-(-197,000 \text{ J/mol})}{(8.314 \text{ J/(mol}\cdot\text{K)})(298 \text{ K})}} = e^{79.5} = 3 \times 10^{34}$$

Part E

2 point(s) maximum

1 point is earned for calculating the number of moles of O₂ needed.

$$PV = nRT$$

$$n = \frac{PV}{RT} = \frac{(0.988 \text{ atm})(0.0400 \text{ L})}{(0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1})(298 \text{ K})} = 0.00162 \text{ mol O}_2$$

1 point is earned for calculating the volume of H₂O₂ solution that should be added.

$$0.00162 \text{ mol O}_2 \times \frac{1 \text{ mol H}_2\text{O}_2}{1 \text{ mol O}_2} = 0.00162 \text{ mol H}_2\text{O}_2 \text{ needed}$$

$$0.00162 \text{ mol H}_2\text{O}_2 \times \frac{\text{L}}{0.800 \text{ mol H}_2\text{O}_2} = 0.00202 \text{ L}$$

Chapter 10 Part 1: Keq



0	1	2
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Student response earns 2 of the following 2 point(s)

2 point(s) maximum

1 point is earned for calculating the number of moles of O₂ needed.

$$PV = nRT$$

$$n = \frac{PV}{RT} = \frac{(0.988 \text{ atm})(0.0400 \text{ L})}{(0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1})(298 \text{ K})} = 0.00162 \text{ mol O}_2$$

1 point is earned for calculating the volume of H₂O₂ solution that should be added.

$$0.00162 \text{ mol O}_2 \times \frac{1 \text{ mol H}_2\text{O}_2}{1 \text{ mol O}_2} = 0.00162 \text{ mol H}_2\text{O}_2 \text{ needed}$$

$$0.00162 \text{ mol H}_2\text{O}_2 \times \frac{\text{L}}{0.800 \text{ mol H}_2\text{O}_2} = 0.00202 \text{ L}$$

Part F

3 point(s) maximum

Part (i)

1 point is earned for the correct reading of the meniscus level to **three** significant figures.

$$36.5 \text{ mL (values within } \pm 0.4 \text{ of 36.5 are acceptable)}$$

Part(ii)

1 point is earned for the correct percent yield.

$$\text{percent yield} = \text{actual yield} / \text{theoretical yield} = 36.5 \text{ mL} / 40.0 \text{ mL} \times 100 = 91.3\%$$

Part(iii)

1 point is earned for the correct answer with a valid explanation. (Only **one** of the two extra gases is required for the point.)

No, the gas also contains water vapor and air that was originally in the flask.

Chapter 10 Part 1: Keq



0	1	2	3
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Student response earns 3 of the following 3 point(s)

3 point(s) maximum

Part (i)

1 point is earned for the correct reading of the meniscus level to **three** significant figures.

36.5 mL (values within ± 0.4 of 36.5 are acceptable)

Part(ii)

1 point is earned for the correct percent yield.

percent yield = actual yield / theoretical yield = 36.5 mL/40.0 mL $\times$ 100 = 91.3%

Part(iii)

1 point is earned for the correct answer with a valid explanation. (Only **one** of the two extra gases is required for the point.)

No, the gas also contains water vapor and air that was originally in the flask.

Part G

1 point(s) maximum

1 point is earned for an appropriate conclusion and valid justification based on the value of K in part (d).

Disagree. The very large value of K implies that the reaction goes essentially to completion, so essentially all of the H_2O_2 reacts to form O_2 .



0	1
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Student response earns 1 of the following 1 point(s)

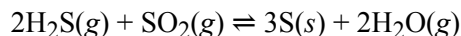
1 point(s) maximum

1 point is earned for an appropriate conclusion and valid justification based on the value of K in part (d).

Disagree. The very large value of K implies that the reaction goes essentially to completion, so essentially all of the H_2O_2 reacts to form O_2 .

Chapter 10 Part 1: Keq

66.



At 298 K, the standard enthalpy change, ΔH° , for the reaction represented above is -145 kilojoules.

(a) Predict the sign of the standard entropy change, ΔS° , for the reaction. Explain the basis for your prediction.

(b) At 298 K, the forward reaction (i.e., toward the right) is spontaneous. What change, if any, would occur in the value of ΔG° for this reaction as the temperature is increased? Explain your reasoning using thermodynamic principles.

(c) What change, if any, would occur in the value of the equilibrium constant, K_{eq} , for the situation described in (b)? Explain your reasoning.

(d) The absolute temperature at which the forward reaction becomes nonspontaneous can be predicted. Write the equation that is used to make the prediction. Why does this equation predict only an approximate value for the temperature?

Part A

1 point is earned for: Statement that ΔS° is negative **1 point is earned for:** 3 moles of gas $\rightarrow$ 2 moles of gas plus solid, (3 moles $\rightarrow$ 2 moles earns no points) OR, 2 gases $\rightarrow$ 1 gas + solid.

OR, use of $\Delta G^\circ - \Delta H^\circ - T\Delta S^\circ$ with $\Delta G^\circ \rightarrow 0$



0	1	2
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The student response earns two of the following points:

1 point is earned for: Statement that ΔS° is negative

1 point is earned for: 3 moles of gas $\rightarrow$ 2 moles of gas plus solid, (3 moles $\rightarrow$ 2 moles earns no points) OR, 2 gases $\rightarrow$ 1 gas + solid.

OR, use of $\Delta G^\circ - \Delta H^\circ - T\Delta S^\circ$ with $\Delta G^\circ \rightarrow 0$

Part B

1 point is earned for: ΔG° is less negative, goes to 0, goes +, gets larger

1 point is earned for: Explanation using $\Delta G^\circ - \Delta H^\circ - T\Delta S^\circ$



0	1	2
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Chapter 10 Part 1: K_{eq}

The student response earns two of the following points:

1 point is earned for: ΔG^0 is less negative, goes to 0, goes +, gets larger

1 point is earned for: Explanation using $\Delta G^0 = \Delta H^0 - T\Delta S^0$

Part C

1 point is earned for: K_{eq} decreases (exponent → more negative) as T increases

OR,

K_{eq} goes from > 1, to 1, to <1 as T increases

1 point is earned for: Correct explanation using the equation $\Delta G^0 = -RT \ln K_{eq}$

Or,

higher T favors the reverse reaction (Le Chatelier) because the forward reaction is exothermic



0	1	2
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The student response earns two of the following points:

1 point is earned for: K_{eq} decreases (exponent → more negative) as T increases

OR,

K_{eq} goes from > 1, to 1, to <1 as T increases

1 point is earned for: Correct explanation using the equation $\Delta G^0 = -RT \ln K_{eq}$

Or,

higher T favors the reverse reaction (Le Chatelier) because the forward reaction is exothermic

Part D

1 point is earned for: Since $\Delta G^0 = 0$ at the point, the equation is $T = \Delta H^0 / \Delta S^0$

($\Delta G^0 = \Delta H^0 - T\Delta S^0$ is not sufficient without $\Delta G^0 = 0$)

1 point is earned for: Prediction is not exact because ΔH^0 and or ΔS^0 vary with T



0	1	2
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Chapter 10 Part 1: Keq

The student response earns two of the following points:

1 point is earned for: Since $\Delta G^0 = 0$ at the point, the equation is $T = \Delta H^0 / \Delta S^0$

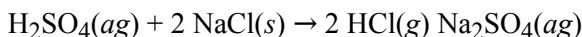
($\Delta G^0 = \Delta H^0 - T\Delta S^0$ is not sufficient without $\Delta G^0 = 0$)

1 point is earned for: Prediction is not exact because ΔH^0 and or ΔS^0 vary with T

Chapter 10 Part 1: Keq

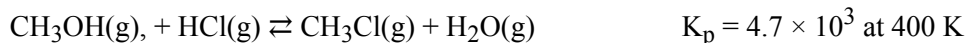
67. Answer the following questions relating to HCl, CH₃Cl, and CH₃Br.

- a. HCl(g), can be prepared by the reaction of concentrated H₂SO₄(aq), with NaCl(s), as represented by the following equation.



- A student claims that the reaction is a redox reaction. Is the student correct? Justify your answer.
- Calculate the mass, in grams, of NaCl(s), needed to react with excess H₂SO₄(aq) to produce 3.0 grams of HCl(g). Assume that the reaction goes to completion.

HCl(g), can react with methanol vapor, CH₃OH(g), to produce CH₃Cl(g), as represented by the following equation.



- CH₃OH(g), and HCl(g), are combined in a 10.0 liter sealed reaction vessel and allowed to reach equilibrium at 400 K. The initial partial pressure of CH₃OH(g) in the vessel is 0.250 atm and that of HCl(g) is 0.600 atm.
 - Does the total pressure in the vessel increase, decrease, or remain the same as equilibrium is approached? Justify your answer in terms of the reaction stoichiometry.
 - Considering the value of K_p , calculate the final partial pressure of HCl(g) after the system inside the vessel reaches equilibrium at 400 K.
 - The student claims that the final partial pressure of CH₃OH(g) at equilibrium is very small but not exactly zero. Do you agree or disagree with the student's claim? Justify your answer.
- The table below shows some data for the compounds CH₃Cl and CH₃Br.

Compound	Molar Mass (g/mol)	Dipole Moment (D)	Boiling Point (K)	$\Delta H_{\text{vaporization}}^\circ$ (kJ/mol)
CH ₃ Cl	50.5	1.87	249	18.9
CH ₃ Br	94.9	1.81	277	22.8

- Identify all the types of intermolecular forces that exist among molecules in CH₃Cl(l).
- In terms of intermolecular forces, explain why the boiling point of CH₃Br(l) is greater than that

Chapter 10 Part 1: Keq

of $\text{CH}_3\text{Cl}(l)$.

- d. A 2.00 milliliter sealed glass vial containing a 1.00 gram sample of $\text{CH}_3\text{Cl}(l)$ is stored in a freezer at 233 K.
- Calculate the pressure in the vial at 298 K assuming that all the $\text{CH}_3\text{Cl}(l)$ vaporizes.
 - Explain why it would be unsafe to remove the vial from the freezer and leave it on a lab bench at 298 K.

Part A

Part i

1 point is earned for indicating the student is incorrect and for providing a valid justification.

No, the student is not correct. None of the oxidation numbers of the elements change ($\text{H} = +1$, $\text{S} = +6$, $\text{O} = -2$, $\text{Na} = +1$, $\text{Cl} = -1$).

Part ii

1 point is earned for determining the number of moles of $\text{HCl}(g)$ (may be implicit).

1 point is earned for calculating a mass of NaCl based on the number of moles of $\text{HCl}(g)$.

$$3.00\text{gHCl} \times \frac{1\text{molHCl}}{36.46\text{gHCl}} \times \frac{2\text{molNaCl}}{2\text{molHCl}} \times \frac{58.44\text{gNaCl}}{1\text{molNaCl}} \\ = 4.81\text{gNaCl}$$



0	1	2	3
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Student earns 3/3 point(s)

Part i

1 point is earned for indicating the student is incorrect and for providing a valid justification.

No, the student is not correct. None of the oxidation numbers of the elements change ($\text{H} = +1$, $\text{S} = +6$, $\text{O} = -2$, $\text{Na} = +1$, $\text{Cl} = -1$).

Part ii

1 point is earned for determining the number of moles of $\text{HCl}(g)$ (may be implicit).

1 point is earned for calculating a mass of NaCl based on the number of moles of $\text{HCl}(g)$.

Chapter 10 Part 1: Keq

$$3.00\text{gHCl} \times \frac{1\text{molHCl}}{36.46\text{gHCl}} \times \frac{2\text{molNaCl}}{2\text{molHCl}} \times \frac{58.44\text{gNaCl}}{1\text{molNaCl}} \\ = 4.81\text{gNaCl}$$

Part B

Part i

1 point is earned for the correct answer and a valid justification.

The pressure will remain the same. The reaction stoichiometry shows that two moles of gaseous reactants produce two moles of gaseous products. Because the number of moles of gas does not change, the pressure does not change.

Part ii

1 point is earned for the correct answer.

The value of K_p is large, so the reaction will proceed to the right until the limiting reactant is essentially used up. Thus practically all of the $\text{CH}_3\text{OH}(g)$ will react and the final pressure of $\text{HCl}(g)$ is $0.600 - 0.250 = 0.350$ atm.

OR

	$\text{CH}_3\text{OH}(g)$	$\text{HCl}(g)$	$\text{CH}_3\text{Cl}(g)$	$\text{H}_2\text{O}(g)$
I	0.250 atm	0.600 atm	0	0
C	$-x$	$-x$	$+x$	$+x$
E	$0.250 - x \approx 0$	$0.600 - x \approx 0.350$	$+x \approx 0.250$	$+x \approx 0.250$

The final pressure of $\text{HCl}(g)$ is 0.350 atm at equilibrium.

Part iii

1 point is earned for the correct choice and a valid justification.

Agree. The large value of K_p means that the partial pressure of the limiting reactant at equilibrium will be extremely small, but some CH_3OH molecules must exist for the system to be in dynamic equilibrium.

OR

$$K_p = \frac{p_{\text{CH}_3\text{Cl}}p_{\text{H}_2\text{O}}}{p_{\text{CH}_3\text{OH}}p_{\text{HCl}}} = 4.7 \times 10^3 = \frac{(0.250)(0.250)}{(p_{\text{CH}_3\text{OH}})(0.350)} \Rightarrow p_{\text{CH}_3\text{OH}} = 3.8 \times 10^{-5}\text{atm}$$

The partial pressure of $\text{CH}_3\text{OH}(g)$ is very small but is not zero.

Chapter 10 Part 1: Keq



0	1	2	3
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Student earns 3/3 point(s)

Part i

1 point is earned for the correct answer and a valid justification.

The pressure will remain the same. The reaction stoichiometry shows that two moles of gaseous reactants produce two moles of gaseous products. Because the number of moles of gas does not change, the pressure does not change.

Part ii

1 point is earned for the correct answer.

The value of K_p is large, so the reaction will proceed to the right until the limiting reactant is essentially used up. Thus practically all of the $\text{CH}_3\text{OH}(g)$ will react and the final pressure of $\text{HCl}(g)$ is $0.600 - 0.250 = 0.350$ atm.

OR

	$\text{CH}_3\text{OH}(g)$	$\text{HCl}(g)$	$\text{CH}_3\text{Cl}(g)$	$\text{H}_2\text{O}(g)$
I	0.250 atm	0.600 atm	0	0
C	$-x$	$-x$	$+x$	$+x$
E	$0.250 - x \approx 0$	$0.600 - x \approx 0.350$	$+x \approx 0.250$	$+x \approx 0.250$

The final pressure of $\text{HCl}(g)$ is 0.350 atm at equilibrium.

Part iii

1 point is earned for the correct choice and a valid justification.

Agree. The large value of K_p means that the partial pressure of the limiting reactant at equilibrium will be extremely small, but some CH_3OH molecules must exist for the system to be in dynamic equilibrium.

OR

$$K_p = \frac{p_{\text{CH}_3\text{Cl}}p_{\text{H}_2\text{O}}}{p_{\text{CH}_3\text{OH}}p_{\text{HCl}}} = 4.7 \times 10^3 = \frac{(0.250)(0.250)}{(p_{\text{CH}_3\text{OH}})(0.350)} \Rightarrow p_{\text{CH}_3\text{OH}} = 3.8 \times 10^{-5} \text{ atm}$$

The partial pressure of $\text{CH}_3\text{OH}(g)$ is very small but is not zero.

Part C

Chapter 10 Part 1: Keq

Part i

1 point is earned for identifying both types of forces.

London dispersion forces and dipole-dipole forces

Part ii

1 point is earned for a valid explanation.

The electron cloud in CH₃Br is larger and more polarizable than that of CH₃Cl. As a result the London dispersion forces are stronger in CH₃Br compared to those in CH₃Cl and consequently the boiling point of CH₃Br is higher than that of CH₃Cl .



0	1	2
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Student earns 2/2 point(s)

Part i

1 point is earned for identifying both types of forces.

London dispersion forces and dipole-dipole forces

Part ii

1 point is earned for a valid explanation.

The electron cloud in CH₃Br is larger and more polarizable than that of CH₃Cl. As a result the London dispersion forces are stronger in CH₃Br compared to those in CH₃Cl and consequently the boiling point of CH₃Br is higher than that of CH₃Cl .

Part D

Part i

1 point is earned for the correct pressure with the correct units.

$$PV = nRT$$

$$n = \frac{1.00\text{gCH}_3\text{Cl}}{50.5\text{g/molCH}_3\text{Cl}} = 0.0198\text{mol}$$

$$P = \frac{nRT}{V} = \frac{(0.0198\text{mol})(0.0821\text{Latmmol}^{-1}\text{K}^{-1})(298\text{K})}{0.00200\text{L}}$$

$$= 242 \text{ atm}$$

Part ii

1 point is earned for a valid explanation.

Chapter 10 Part 1: Keq

At room temperature the liquid will vaporize. Consequently the glass vial may not be strong enough to withstand the increase in pressure.



0	1	2
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Student earns 2/2 point(s)

Part i

1 point is earned for the correct pressure with the correct units.

$$PV = nRT$$

$$n = \frac{1.00\text{gCH}_3\text{Cl}}{50.5\text{g/molCH}_3\text{Cl}} = 0.0198\text{mol}$$

$$P = \frac{nRT}{V} = \frac{(0.0198\text{mol})(0.0821\text{Latmmol}^{-1}\text{K}^{-1})(298\text{K})}{0.00200\text{L}}$$

$$= 242 \text{ atm}$$

Part ii

1 point is earned for a valid explanation.

At room temperature the liquid will vaporize. Consequently the glass vial may not be strong enough to withstand the increase in pressure.

Chapter 10 Part 1: Keq

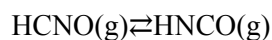
68. Answer the following questions about the isomers fulminic acid and isocyanic acid.

Two possible Lewis electron-dot diagrams for fulminic acid, HCNO, are shown below.



- a. Explain why the diagram on the left is the better representation for the bonding in fulminic acid. Justify your choice based on formal charges.

Fulminic acid can convert to isocyanic acid according to the equation below.



fulminic acid - isocyanic acid

Fulminic Acid	Isocyanic Acid
$\text{H}-\text{C}\equiv\text{N}-\ddot{\text{O}}:$	$\text{H}-\ddot{\text{N}}=\text{C}=\ddot{\text{O}}:$

- b. Using the Lewis electron-dot diagrams of fulminic acid and isocyanic acid shown in the boxes above and the table of average bond enthalpies below, determine the value of ΔH° for the reaction of HCNO(g) to

form HNCO(g).	Bond	Enthalpy (kJ/mol)		Bond	Enthalpy (kJ/mol)		Bond	Enthalpy (kJ/mol)
	N-O	201		C=N	615		H-C	413
	C=O	745		C≡N	891		H-N	391

- c. A student claims that ΔS° for the reaction is close to zero. Explain why the student's claim is accurate.
- d. Which species, fulminic acid (HCNO) or isocyanic acid (HNCO), is present in higher concentration at equilibrium at 298 K? Justify your answer in terms of thermodynamic favorability and the equilibrium constant.

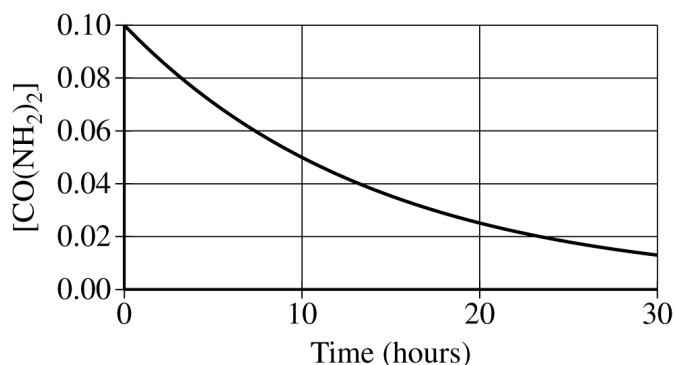
The ammonium salt of isocyanic acid is a product of the decomposition of urea, $\text{CO}(\text{NH}_2)_2$, represented below.



A student studying the decomposition reaction runs the reaction at 90°C. The student collects data on the concentration of urea as a function of time, as shown by the data table and the graph below.

Chapter 10 Part 1: Keq

Time (hours)	[CO(NH ₂) ₂]
0	0.1000
5	0.0707
10	0.0500
15	0.0354
20	0.0250
25	0.0177
30	0.0125



- e. The student proposes that the rate law is $rate = k[CO(NH_2)_2]$.
- Explain how the data support the student's proposed rate law.
 - Using the proposed rate law and the student's results, determine the value of the rate constant, k . Include units with your answer.
- f. The student learns that the decomposition reaction was run in a solution with a pH of 13. Briefly describe an experiment, including the initial conditions that you would change and the data you would gather, to determine whether the rate of the reaction depends on the concentration of $OH^-(aq)$.

Part A

2 point(s) maximum

1 point is earned for the correct assignment of formal charges in the two diagrams.

In the diagram on the left, the C atom has a formal charge of zero and the O atom has a formal charge of -1. In the diagram on the right, the C atom has a formal charge of -1 and the O atom has a formal charge of zero.

1 point is earned for a correct explanation.

The diagram on the left is the better representation because it puts the negative formal charge on oxygen, which is more electronegative than carbon.



0	1	2
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Student response earns 2 of the following 2 point(s)

2 point(s) maximum

1 point is earned for the correct assignment of formal charges in the two diagrams.

In the diagram on the left, the C atom has a formal charge of zero and the O atom has a formal charge of -1. In the diagram on the right, the C atom has a formal charge of -1 and the O atom has a formal charge of zero.

Chapter 10 Part 1: Keq

1 point is earned for a correct explanation.

The diagram on the left is the better representation because it puts the negative formal charge on oxygen, which is more electronegative than carbon.

Part B

2 point(s) maximum

1 point is earned for subtracting the enthalpies of bonds formed from the enthalpies of bonds broken.

1 point is earned for the correct determination of ΔH° .

Compound	HCNO	HNCO
Bond Enthalpies (kJ/mol)	413 + 891 + 201	391 + 615 + 745
Total Bond Enthalpy (kJ/mol)	1505	1751

$$\begin{aligned}\Delta H^\circ &= \Sigma(\text{enthalpies of bonds broken}) - \Sigma(\text{enthalpies of bonds formed}) \\ &= 1505 \text{ kJ/mol} - 1751 \text{ kJ/mol} \\ &= -246 \text{ kJ/mol}_{\text{rxn}}\end{aligned}$$



0	1	2
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Student response earns 2 of the following 2 point(s)

2 point(s) maximum

1 point is earned for subtracting the enthalpies of bonds formed from the enthalpies of bonds broken.

1 point is earned for the correct determination of ΔH° .

Compound	HCNO	HNCO
Bond Enthalpies (kJ/mol)	413 + 891 + 201	391 + 615 + 745
Total Bond Enthalpy (kJ/mol)	1505	1751

$$\begin{aligned}\Delta H^\circ &= \Sigma(\text{enthalpies of bonds broken}) - \Sigma(\text{enthalpies of bonds formed}) \\ &= 1505 \text{ kJ/mol} - 1751 \text{ kJ/mol} \\ &= -246 \text{ kJ/mol}_{\text{rxn}}\end{aligned}$$

Part C

1 point(s) maximum

1 point is earned for a correct explanation.

The change from fulminic acid to isocyanic acid is a rearrangement of atoms with no change in phase or number of molecules.

Chapter 10 Part 1: Keq



0	1
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Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for a correct explanation.

The change from fulminic acid to isocyanic acid is a rearrangement of atoms with no change in phase or number of molecules.

Part D

2 point(s) maximum

1 point is earned for the correct choice with a valid justification. (Calculation of ΔG° is a sufficient justification.)

Isocyanic acid (HNCO) will be present in higher concentration.

ΔG° is essentially equal to ΔH° because ΔS° is essentially zero, so $\Delta G^\circ \approx -246 \text{ kJ/molrxn}$, indicating the forward reaction is thermodynamically favorable.

1 point is earned for correctly connecting thermodynamic favorability to the equilibrium constant, K .

Since ΔG° is negative, $K > 1$ ($\Delta G^\circ = -RT \ln K$), resulting in a higher concentration of product than reactant at equilibrium.



0	1	2
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Student response earns 2 of the following 2 point(s)

2 point(s) maximum

1 point is earned for the correct choice with a valid justification. (Calculation of ΔG° is a sufficient justification.)

Isocyanic acid (HNCO) will be present in higher concentration.

ΔG° is essentially equal to ΔH° because ΔS° is essentially zero, so $\Delta G^\circ \approx -246 \text{ kJ/molrxn}$, indicating the forward reaction is thermodynamically favorable.

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Chapter 10 Part 1: Keq**Part E****2 point(s) maximum**

Part(i)

1 point is earned for a correct explanation.

From inspecting the data table or the graph, it is evident that the decomposition reaction has a constant half-life, which indicates that the reaction is a first-order reaction.

Part (ii)

1 point is earned for the correct value of k with correct units.

Since the reaction is first order,

$$k = 0.693 / t_{1/2} = 0.693 / 10.\text{h} = 0.069 \text{ h}^{-1}$$

or

$$k = (\ln[A]_0 - \ln[A]_t) / t = (\ln(0.1000) - \ln(0.0500)) / 10.\text{h} = 0.069 \text{ h}^{-1}$$



0	1	2
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Student response earns 2 of the following 2 point(s)**2 point(s) maximum**

Part(i)

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Part F

Chapter 10 Part 1: Keq

1 point(s) maximum

1 point is earned for the description of a valid experiment.

Perform the experiment at a different concentration of $\text{OH}^-(aq)$ and measure how the concentration of $\text{CO}(\text{NH}_2)_2$ changes over time. (Other variables, such as temperature, should be held constant.)



0	1
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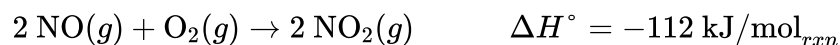
Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for the description of a valid experiment.

Perform the experiment at a different concentration of $\text{OH}^-(aq)$ and measure how the concentration of $\text{CO}(\text{NH}_2)_2$ changes over time. (Other variables, such as temperature, should be held constant.)

69. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Answer the following questions about the reaction represented above at 298 K.

	Absolute Entropy at 298 K (J/(K mol))
$\text{NO}(g)$	211
$\text{O}_2(g)$	205
$\text{NO}_2(g)$	240

- (a) Using the information in the table above, calculate the value of ΔS° for the reaction.
- (b) Is the reaction thermodynamically favorable at 298 K? Justify your answer based on the calculation of the value and sign of ΔG° .
- (c) Is the value of the equilibrium constant, K , for the reaction greater than 1, or less than 1? Justify your answer.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 1: Keq



0	1
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The response gives the both the following set up and calculation:

$$\Delta S^\circ = \Sigma S^\circ \text{ products} - \Sigma S^\circ \text{ reactants}$$

$$\Delta S^\circ = 2(240) - [2(211) + 205] = -147 \text{ J}/(\text{K mol}_{rxn})$$

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets both of the following criteria:

- ☐ The response includes the following calculation.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta G^\circ = -112 \text{ kJ/mol}_{rxn} - (298 \text{ K})(-0.147 \text{ kJ}/(\text{K mol}_{rxn})) = -68 \text{ kJ/mol}_{rxn}$$
- ☐ The response includes the following conclusion: The reaction is thermodynamically favorable because ΔG° is negative.

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. The answer to part (d) should be consistent with the answer to part (c).



0	1
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The response indicates that the equilibrium constant is greater than 1 because $\Delta G^\circ < 0$

Chapter 10 Part 1: Keq

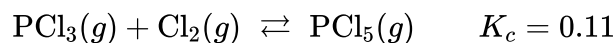
70. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to PCl_3 .

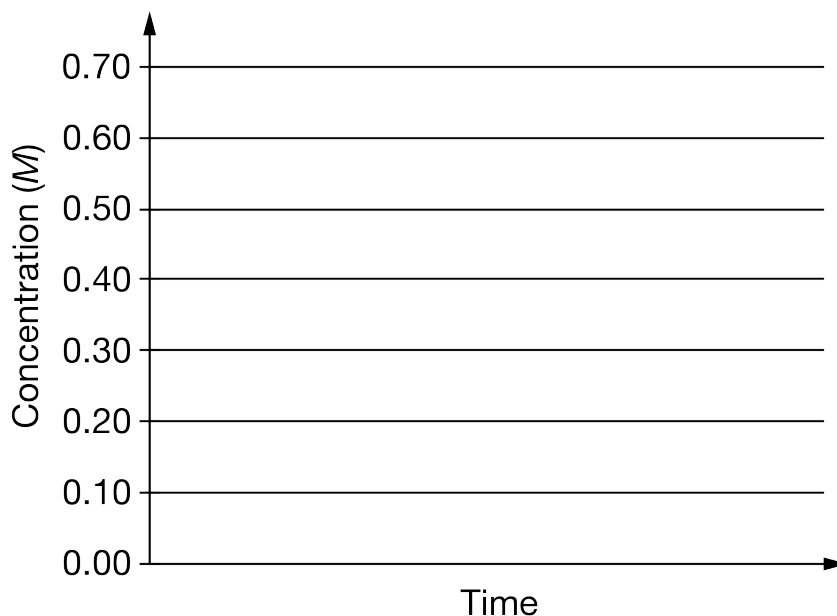
- (a) In the box below, draw the complete Lewis electron-dot diagram for PCl_3 .



- (b) What is the shape of the PCl_3 molecule?



A 0.60 mol sample of $\text{PCl}_3(g)$ and a 0.70 mol sample of $\text{Cl}_2(g)$ are placed in a previously evacuated 1.0 L rigid container, and the reaction represented above takes place. At equilibrium, the concentration of $\text{PCl}_5(g)$ in the container is 0.040 M.



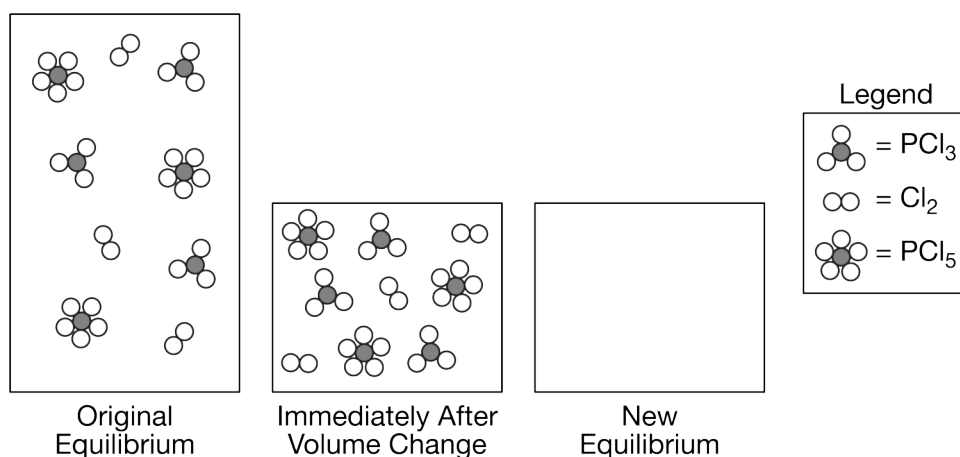
Chapter 10 Part 1: Keq

(c) On the graph above, carefully draw three curves, one for each of the three gases. The curves must show how the concentration of each of the three gases changes as equilibrium is established. Label each curve with the formula of the gas.

(d) Explain how the slopes of the curves at equilibrium can be used to show that the rates of the forward and reverse reactions are the same at equilibrium.

(e) As the reaction occurs at constant temperature, does the pressure inside the container increase, decrease, or remain the same? Explain.

Another mixture of $\text{PCl}_3(g)$, $\text{Cl}_2(g)$, and $\text{PCl}_5(g)$ is at equilibrium at a different temperature in a different vessel. The system at equilibrium is represented by the following box on the left. Then the volume of the vessel is decreased. The box in the middle represents the system before equilibrium is reestablished at the same temperature.



(f) In the box above on the right, draw an appropriate number of each type of molecule to represent a possible new equilibrium.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response meets both of the following criteria:

- ☐ There is a P atom connected to three Cl atoms by single bonds.
- ☐ There are three nonbonding pairs of electrons on each Cl atom and one on the P atom.

Part (b)

Chapter 10 Part 1: Keq

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. The response must be consistent with the answer to part (a). For example, if the student does not put one pair of nonbonding electrons on the P atom, the consistent response that would earn the point is trigonal planar.



0	1
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The response indicates that the molecule is trigonal pyramidal.

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The response meets all of the following criteria:

- ☐ The curve for PCl_3 starts at 0.60 M and levels off at 0.56 M
- ☐ The curve for Cl_2 starts at 0.70 M and levels off at 0.66 M .
- ☐ The curve for PCl_5 starts at 0.00 M and levels off at 0.040 M .

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the curves are horizontal (slope = 0) at equilibrium, so the concentrations of the reactants and product are not changing, therefore the rate of the forward reaction must be the same as the rate of the reverse reaction.

Part (e)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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Chapter 10 Part 1: Keq

The response meets both of the following criteria:

- ☐ The response indicates that the pressure decreases.
- ☐ The response indicates that there are more moles of reactant gases than product gas.

Part (f)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets both of the following criteria:

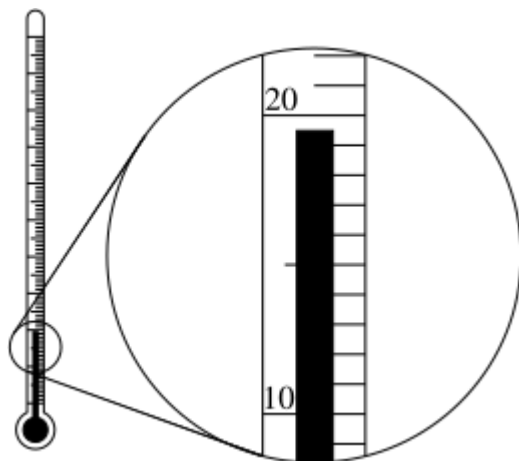
- ☐ There are more than three PCl_5 molecules and fewer than three PCl_3 and fewer than three Cl_2 molecules drawn in the box.
- ☐ There is conservation of atoms (six P atoms and thirty Cl atoms).

Chapter 10 Part 1: Keq

71. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

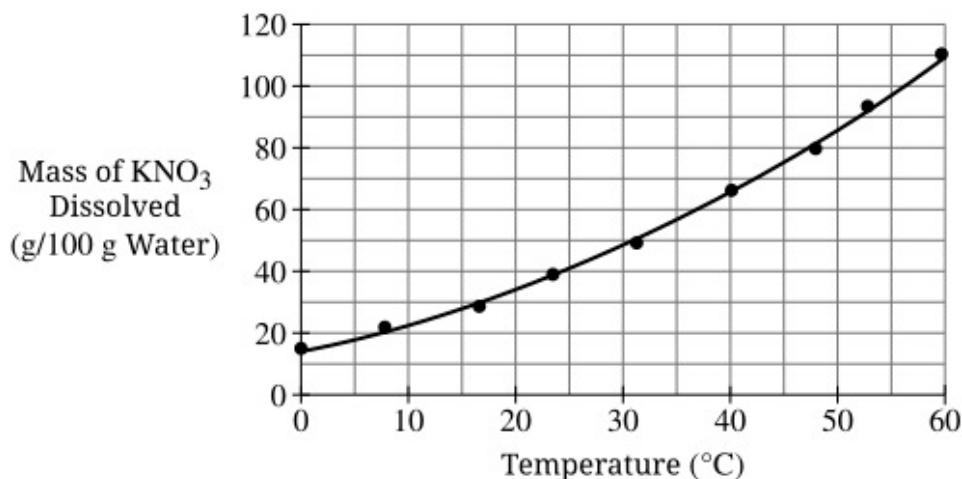
A student conducts a solubility experiment with potassium nitrate, KNO_3 , in water by identifying the temperature at which various masses of KNO_3 will completely dissolve.

(a) At room temperature, H_2O is a liquid and KNO_3 is a solid. The student incorrectly claims that this is because potassium and nitrate ions experience stronger dispersion forces than H_2O molecules. In terms of the types of attractive forces, explain why the student is incorrect.



- (b) In one of the trials, the student reads the initial temperature of the water using the Celsius thermometer above. Record this temperature to the appropriate number of significant figures.
- (c) At the temperature indicated on the thermometer, the student adds the solid KNO_3 to the sample of water and begins to stir the mixture. Explain the effect of stirring on the maximum amount of solid that can dissolve in the water.

Chapter 10 Part 1: Keq



The graph above summarizes the data collected by the student.

(d) Another student places 38.1 grams of KNO_3 into 100.0 g of water at the temperature measured in part (b), stirs the mixture, and observes that the solid does not dissolve completely. The student plans to heat the mixture and states that the solid will completely dissolve before the mixture reaches 30°C . Do you agree with the claim? Justify your answer with information from the graph.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response provides a correct explanation.

- Dispersion forces are weaker than the ionic bonds present in KNO_3 , and dispersion forces are not the dominant attractive force that leads to the properties of ionic compounds; dispersion forces of attraction are more relevant between molecules in molecular compounds.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response provides the correct answer.

Chapter 10 Part 1: Keq

- 19.5 °C

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a correct explanation.

- The stirring will not affect the maximum amount of KNO_3 that will dissolve (which depends on the equilibrium between the solid and the dissolved ions), only the rate at which the solid dissolves.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct answer and justification.

- Agree. According to the graph, the maximum solubility at 30 °C should be approximately 48 g so 38.1 g is not enough solute to make a saturated solution at this temperature, so the solid should completely dissolve.